

What is included to ensure the full reproducibility of the project?

- Project Overview. ☐
- Data Source: a structured table listing each document title, its inclusion decision, and a clear, concise justification for that decision. ☐
- A complete ERD showing all entities, attributes, and relationships. ☐
- Database schema normalized up to the Fifth Normal Form (5NF). ☐
- Summary of the rationale behind the database design, including design decisions, normalization strategy, deviations from normal forms, and the data dictionary. ☐
- Well-documented Python scripts for data cleaning, preprocessing, and SQL generation, designed for reproducibility, clarity, and maintainability. ☐
- SQL scripts for database construction, along with a verified database view on the virtual machine. ☐
- SQL scripts for populating the database with cleaned and structured data, supported by evidence of successful population. ☐
- SQL queries for data retrieval, filtering, and analysis, accompanied by their corresponding outputs. ☐
- A comprehensive graph-based analysis using Neo4j. ☐
- Integrated table of key mutated genes with functional, clinical, and frequency annotations from multiple databases. ☐
- Reproduction steps. ☐
- Limitations and future work. ☐

***1. Project overview***

This project aims to address a central challenge in cancer bioinformatics: the lack of a consistent and integrated structure for multi-omics data. Genomic datasets from platforms such as cBioPortal are distributed across multiple files, each representing a different data modality, including mutation, copy number alteration (CNA), gene expression, and clinical information. These datasets are heterogeneous in structure, can sometimes be inconsistent in identifiers, and are difficult to integrate. As a result, a significant portion of the analysis effort is spent on preprocessing rather than extracting biological insight.

This work resolves that challenge by designing and implementing a fully integrated, gene-centered relational database for Low-Grade Glioma (LGG) using TCGA PanCancer Atlas data. The database is built on a strict design principle: every molecular observation must be linked to both a gene and a sample. This constraint ensures consistency across all data layers, enabling the direct integration of mutation, CNA, and gene expression data within a unified and queryable framework.

A rigorous data curation strategy was applied at both the dataset and variable levels. Only features that provide direct analytical value and align with the gene-centered design were retained. High-level, redundant, or non-compatible data types, such as arm-level CNA, segment-level data, methylation, proteomics, and structural variants, were systematically excluded to maintain clarity, consistency, and interpretability. Clinical variables were carefully selected to preserve essential biological context, including demographics, tumor characteristics, and survival outcomes, while removing administrative and non-informative fields.

The project implements a fully reproducible Python-based pipeline that performs data cleaning, validation, sample-level harmonization, gene-level harmonization, and HGNC-based gene identifier mapping. The pipeline enforces strict consistency by retaining only shared samples and genes across all datasets and standardizing identifiers to eliminate ambiguity. The final output consists of fully matched multi-omics datasets that are clean, non-redundant, and ready for database ingestion.

The database schema is normalized up to the Fifth Normal Form (5NF), ensuring minimal redundancy and strong data integrity. It follows a modular design with separate entities for patients, diagnoses, samples, genes, and each omics

layer. Many-to-many relationships are resolved using bridge tables, allowing accurate representation of complex biological relationships while maintaining scalability and query efficiency.

The final system supports direct interaction through SQL for structured querying and analytical exploration. In addition, a subset of the integrated data is exported for graph-based analysis using Neo4j, where sample-gene relationships are modeled as networks to support visualization and subtype-specific pattern discovery. This graph component is used as an analytical extension and does not represent the full database.

The outcome is a high-integrity, reproducible, and analysis-ready multi-omics database that eliminates the burden of data preprocessing. It enables efficient exploration of mutation patterns, subtype-specific genomic signatures, and gene-level interactions across integrated data layers, allowing researchers to focus on biological interpretation rather than data preparation.

## 2. Data Source

Table 1: Dataset and Variable-Level Screening and Inclusion Assessment Table

| Document                  | Variable/s                                                                                                                                                                                                                                    | Inclusion Decision                                                                  | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data_armlevel_cna.txt     | Full document                                                                                                                                                                                                                                 |    | This file was excluded because it reports copy number changes at the chromosomal arm level, meaning it captures broad gains or losses across large regions such as 1p or 1q rather than changes tied to specific genes. The database in this project is designed to work at the gene level, where each feature can be directly linked to a gene and a sample to support clear and interpretable analysis. Arm-level data does not offer that level of detail, so it cannot be reliably integrated with mutation data or used to assess functional impact at the gene level. Including it would introduce ambiguity and make the overall structure less consistent, especially when more precise gene-level CNA data is better suited for this purpose. For these reasons, the dataset was excluded to keep the analysis focused, consistent, and easier to interpret.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| data_clinical_patient.txt | Patient_Identifier,<br>Diagnosis_Age,<br>Gender, Cancer_Type,<br>Cancer_Subtype,<br>OS_Status, OS_Months,<br>DSS_Status, DSS_Months,<br>DFS_Status, DFS_Months,<br>PFS_Status, PFS_Months,<br>Genetic_Ancestry_Label                          |  | These variables were included to ensure that the genomic dataset is supported by essential clinical and demographic context while keeping the structure clean and focused. Patient_Identifier allows consistent linking across mutation, CNA, and expression data, which is critical for maintaining data integrity. Diagnosis_Age and Gender provide basic biological context and are commonly used as covariates in genomic analyses, especially when exploring variation across patient groups. Cancer_Type and Cancer_Subtype are necessary to capture disease heterogeneity within glioma and to enable more precise, subtype-driven molecular analysis. The survival variables (OS, DSS, DFS, and PFS along with their corresponding time measures) were retained because they allow direct connection between genomic features and clinical outcomes, which is important for any future prognosis-related analysis.<br><br>Genetic_Ancestry_Label was included to account for population-level variation, helping to avoid confounding effects and improving the interpretation of genomic patterns. Overall, these variables were selected because they add clear analytical value without introducing redundancy, and they align well with the gene-centered design of the database while still supporting future clinical integration. |
|                           | Other_Patient_ID, AJCC_Stage_Code,<br>AJCC_Staging_Version,<br>Days_To_Last_Contact_<br>From_Diagnosis,<br>Age_At_Diagnosis_Days,<br>Days_Last_Alive_From_Diagnosis,<br>Ethnicity, Race, Form_Date,<br>Consent_Verified, Neoadjuvant_Therapy, |  | Several variables were excluded during dataset curation because they were either redundant, not aligned with the gene-centric design of the database, or did not provide usable information. Identifiers such as Other_Patient_ID were removed since the primary patient identifier already ensures accurate linking across all data layers. Clinical staging variables including AJCC_Stage_Code, AJCC_T_Stage, AJCC_N_Stage, AJCC_M_Stage, and AJCC_Staging_Version were excluded because they do not directly contribute to mutation, CNA, or gene expression                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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|                          | ICD10_Code, ICD03_Histology, ICD03_Site, New_Neoplasm_Event, AJCC_M_Stage, AJCC_N_Stage, AJCC_T_Stage, Cancer_Status, LN_Presentation_Status, Prior_Diagnosis, Radiation_Therapy, Patient_Weight_kg, PanCan_Pathway_Analysis                                  |   | analysis and would not integrate meaningfully with the molecular data structure. Time-related variables such as Days_To_Last_Contact_From_Diagnosis, Days_Last_Alive_From_Diagnosis, and Age_At_Diagnosis_Days were excluded due to redundancy with already retained survival metrics or because they do not add additional analytical value. Demographic fields like Ethnicity and Race were removed in favor of Genetic_Ancestry_Label, which provides more relevant biological context for genomic interpretation. Administrative variables such as Form_Date and Consent_Verified were not required for analysis. Treatment-related fields including Neoadjuvant_Therapy and Radiation_Therapy, along with broader clinical descriptors such as ICD10_Code, ICD03_Histology, ICD03_Site, Cancer_Status, Prior_Diagnosis, and New_Neoplasm_Event, were excluded because they fall outside the scope of the current genomic-focused integration. In addition, variables such as LN_Presentation_Status and Patient_Weight_kg contained no usable data and were therefore removed. Finally, PanCan_Pathway_Analysis was excluded because it does not represent raw genomic or clinical data but rather a precomputed annotation indicating whether a sample was included in a broader PanCancer pathway analysis. |
| data_clinical_sample.txt | Patient_Identifier, Sample_Identifier, Tumor_Type, Neoplasm_Histologic_Grade, Tumor_Disease_Anatomic_Site, Aneuploidy_Score, Sample_Type, Somatic_Status                                                                                                      | ✓ | These variables were included because they provide the essential sample-level context needed to interpret genomic alterations while keeping the dataset focused and structured. Patient_Identifier and Sample_Identifier are critical for maintaining accurate linkage and traceability across clinical, mutation, CNA, and expression data. Tumor_Type, Neoplasm_Histologic_Grade, and Tumor_Disease_Anatomic_Site were retained because they capture key pathological characteristics that directly influence biological interpretation and support subgroup analysis within glioma. Aneuploidy_Score was included as a relevant summary measure of chromosomal imbalance that complements mutation and CNA data without introducing unnecessary complexity. Sample_Type was kept to distinguish between different specimen categories, which is important for quality control and correct interpretation of molecular results. Somatic_Status was included to confirm that the alterations being analyzed are somatic, which aligns directly with the scope of this project.                                                                                                                                                                                                                                    |
|                          | Oncotree_Code, Cancer_Type, Cancer_Type_Detailed, Tissue_Pro prospective_Collection_Indicator, Tissue_Retrospective_Collection_Indicator, Tissue_Source_Site_Code, MSI_MANTIS_Score, MSIsensor_Score, TMB_Nonsynonymous, Tissue_Source_Site, Tumor_Break_Load | ✗ | These variables were excluded to maintain a clean, non-redundant, and analysis-driven dataset. Fields such as Oncotree_Code, Cancer_Type, and Cancer_Type_Detailed were excluded because the disease classification is already captured using the retained tumor descriptors, making these variables redundant. Collection-related metadata, including Tissue_Pro prospective_Collection_Indicator, Tissue_Retrospective_Collection_Indicator, and Tissue_Source_Site_Code, were removed because they do not contribute to the interpretation of genomic features. Similarly, MSI_MANTIS_Score, MSIsensor_Score, and TMB_Nonsynonymous were excluded because they are not central to glioma biology and are not required for the mutation, CNA, and expression analyses defined in this project. Tumor_Break_Load was also excluded since it reflects structural variation at a broader level and does not integrate directly with the gene-level framework used in this database. Institutional and site-related variables such as Tissue_Source_Site were removed as they do not influence molecular analysis. Overall, these exclusions ensure that the dataset remains focused, avoids redundancy, and aligns with a gene-centered structure optimized for accurate and interpretable genomic analysis.        |
| data_mutations.txt       | Hugo_Symbol, Entrez_Gene_ID, Chromosome, Start_Position,                                                                                                                                                                                                      | ✓ | The mutation dataset was curated to retain only the variables that directly support clear, gene-centered genomic analysis and efficient integration with other data types. Hugo_Symbol and Entrez_Gene_ID were included to uniquely identify each gene, while Chromosome,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|                                | End_Position, Consequence, Variant_Classification, Variant_Type, Reference_Allele, Tumor_Seq_Allele2, Sample_ID, VAF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                     | Start_Position, and End_Position provide precise genomic localization of each variant. Functional impact is captured through Consequence, Variant_Classification, and Variant_Type, which are sufficient to describe the biological effect of each mutation at a level appropriate for downstream analysis. Sequence-level changes are represented using Reference_Allele and Tumor_Seq_Allele2, and Sample_ID ensures correct linkage between each mutation and its corresponding sample. VAF was retained as a compact and interpretable measure of mutation prevalence within the sample. Together, these variables provide a clean and sufficient structure for mutation analysis without unnecessary complexity, while remaining fully compatible with CNA and expression datasets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                | Center, NCBI_Build, Strand, Tumor_Seq_Allele1, dbSNP_RS, dbSNP_Val_Status, Matched_Norm_Sample_Barcode, Match_Norm_Seq_Allele1, Match_Norm_Seq_Allele2, Tumor_Validation_Allele1, Tumor_Validation_Allele2, Match_Norm_Validation_Allele1, Match_Norm_Validation_Allele2, Verification_Status, Validation_Status, Mutation_Status, Sequencing_Phase, Sequence_Source, Validation_Method, Score, BAM_File, t_ref_count, t_alt_count, n_ref_count, n_alt_count, HGVSp, HGVSp_Short, Transcript_ID, RefSeq, Protein_position, Codons, Hotspot, AA_MAF, AFR_MAF, ALLELE_NUM, AMR_MAF, ASN_MAF, Allele, Amino_acids, Biotype, Canonical, CCDS, CDS_position, CENTERS, CLIN_SIG, CONTEXT, Cosmic, DBVS, Distance, Domains, EAS_MAF, EA_MAF, ENSP, EUR_MAF, EXON, ExAC_AF, ExAC_AF_AFR, ExAC_AF_AMR, ExAC_AF_EAS, ExAC_AF_FIN, ExAC_AF_NFE, ExAC_AF_OTH, ExAC_AF_SAS, Existing_variation, Exon_Number, FILTER, Feature, Feature_type, Gene_Pheno, Gmaf, Gene, Hgnc_Id, Hgvs_Offset, High_Inf_Pos, Impact, Intron, Mergesource, Minimised, Motif_Name, Motif_Pos, Motif_Score_Change, Ncallers, Pheno, Pick, Pubmed, Polyphen, SAS_MAF, Sift, Somatic, Swissprot, Symbol, Symbol_Source, Trembl, Tsl, Uniparc, Variant_Class, All_Effects, Cdna_Position, n_Depth, Annotation_Status, t_depth, HGVSc |  | <p>A large set of variables was excluded to keep the mutation dataset focused, consistent, and aligned with a gene-centered analytical framework. Technical and administrative metadata such as Center, NCBI_Build, Strand, BAM_File, Sequencing_Phase, Sequence_Source, Validation_Method, and related fields were removed because they describe how the data was generated rather than its biological meaning. Validation and quality-related variables, including Verification_Status, Validation_Status, Mutation_Status, Score, FILTER, Ncallers, and Annotation_Status, were excluded due to inconsistency and limited value for the intended analyses. Normal sample-related fields and extended allele representations, such as Matched_Norm_Sample_Barcode, Match_Norm_Seq_Allele1, Match_Norm_Seq_Allele2, Tumor_Seq_Allele1, and validation allele fields, were also removed to avoid redundancy and maintain focus on tumor-derived variants.</p> <p>The read count variables t_ref_count and t_alt_count were used specifically to calculate Variant Allele Frequency (VAF), which was retained as a concise and interpretable measure of mutation abundance; after this calculation, these fields, along with other depth-related variables such as n_ref_count, n_alt_count, t_depth, and n_Depth, were excluded to reduce redundancy and dataset size. Protein-level, transcript-specific, and detailed sequence annotation fields such as HGVSp, HGVSp_Short, HGVSc, Transcript_ID, RefSeq, Protein_position, Codons, CDS_position, Cdna_Position, EXON, Intron, Exon_Number, Feature, Feature_type, ENSP, and CCDS were removed to avoid unnecessary complexity that does not add value at the current level of analysis. Functional prediction and annotation variables, including SIFT, PolyPhen, Impact, Biotype, Canonical, Domains, and motif-related features, were excluded because they are analysis-specific and can be incorporated later if needed.</p> <p>Population frequency metrics and external database annotations, such as AA_MAF, AFR_MAF, AMR_MAF, ASN_MAF, EAS_MAF, EUR_MAF, SAS_MAF, ExAC_AF and its subcategories, GMAF, COSMIC-related identifiers, Existing_variation, DBVS, CLIN_SIG, and PubMed references, were also removed because they are not required for constructing the core mutation dataset and may introduce sparsity. Additional cross-referencing and descriptive fields, including Gene, Hgnc_Id, Symbol, Symbol_Source, Swissprot, Trembl, Uniparc, Variant_Class, All_Effects, Allele, Amino_acids, Distance, High_Inf_Pos, Gene_Pheno, Pheno, Pick, and Tsl, were excluded to prevent duplication and maintain a clean structure. Overall, these exclusions eliminate redundancy and excessive detail while preserving all essential information needed for accurate, efficient, and interpretable genomic analysis.</p> |
| data_clinical_supp_hypoxia.txt | Buffa_Hypoxia_Score, Winter_Hypoxia_Score, Ragnum_Hypoxia_Score                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  | The hypoxia dataset was included because it adds a functional layer derived from mRNA expression that complements the genomic data without introducing unnecessary complexity. The Buffa, Winter, and Ragnum hypoxia scores are well-established gene expression-based signatures that summarize tumor hypoxic status in a compact and interpretable form. This is particularly important in glioma, where                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

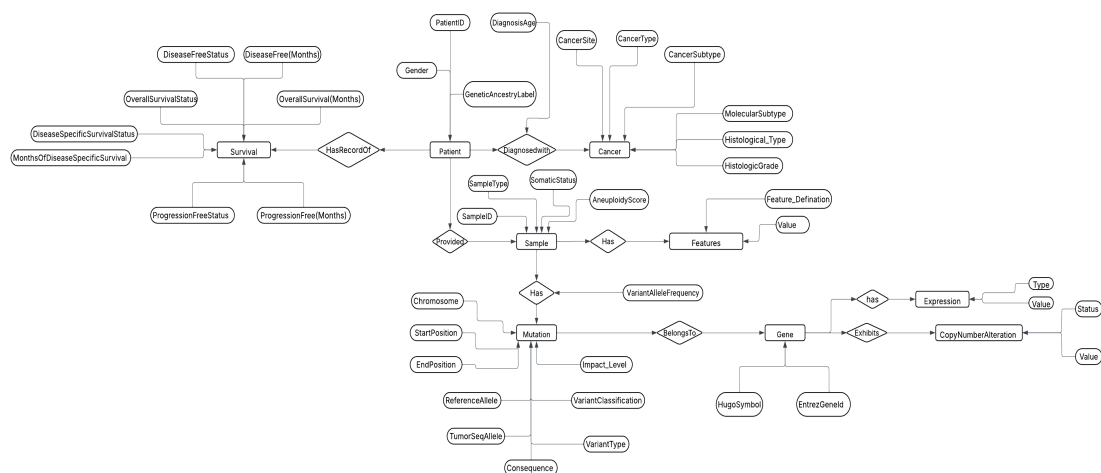
|                            |               |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|                            |               |   | hypoxia is closely linked to tumor aggressiveness, progression, and response to therapy. Including these scores allows direct integration between molecular alterations such as mutations and CNA and the tumor microenvironment, which improves the biological interpretation of the dataset and supports more meaningful downstream analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| data_cna_hg19.seg          | Full Document | ✗ | This file was excluded because it contains raw segmented copy number data that is not directly compatible with the structured, gene-centered design of the database. The file includes low-level segment information such as chromosomal regions, genomic coordinates, probe counts, and segment mean values, which require extensive preprocessing and transformation before they can be used in analysis. In its current form, the data is highly granular and organized at the segment level rather than the gene or sample level, which makes it difficult to integrate with mutation and expression datasets. Including it would increase complexity, introduce redundancy across overlapping segments, and reduce the clarity of downstream analysis. For this reason, it was excluded in favor of higher-level CNA representations that can be directly linked to genes and samples in a consistent and interpretable way.                       |
| data_cna.txt               | Full document | ✓ | The transformed CNA dataset was included because it provides a clean, gene-level representation of copy number alterations that can be directly integrated with mutation and expression data. The original segment-based data was transformed into a gene-by-sample structure, where each record is tied to a specific gene and sample. This aligns fully with the gene-centered design of the database and allows straightforward interpretation of copy number states across the cohort. Hugo_Symbol and Entrez_Gene_Id ensure consistent gene identification, while Sample_ID enables proper linkage across all data layers. CNA_Value captures the magnitude of copy number change, and CNA_Status provides an interpretable categorical representation of gain or loss.                                                                                                                                                                            |
| data_gene_panel_matrix.txt | Full document | ✗ | The data_gene_panel_matrix file was excluded because it does not contain quantitative or biologically informative genomic data. The variables it includes, such as mutations, gistic, and structural_variants, only indicate the sequencing platform or data availability (for example, WXS) rather than actual measured values or genomic alterations. As a result, the file provides descriptive metadata rather than features that can be used in analysis. Including it would not contribute to mutation, CNA, or expression studies and would add unnecessary redundancy, since the presence of sequencing data is already reflected in the main datasets. For this reason, it was excluded to maintain a focused, analysis-ready database built on meaningful and directly interpretable genomic variables.                                                                                                                                       |
| data_genetic_ancestry.txt  | Full document | ✗ | The genetic ancestry file was excluded because it contains admixture proportions such as EUR, AFR, EAS, and AMR, which describe population background rather than direct genomic features. These variables are useful for population-level stratification, but they do not represent mutations, copy number alterations, or gene expression, and therefore do not align with the core gene-centered design of this database. Including them would add an additional layer of variables that cannot be directly linked to genes or molecular events, increasing complexity without improving the interpretation of mutation or CNA data. Since a simplified genetic ancestry label was already retained in the clinical dataset to capture population context, this file was considered redundant. For these reasons, it was excluded to maintain a focused, consistent, and analysis-ready dataset centered on directly interpretable genomic features. |
| data_log2_cna.txt          | Full document | ✗ | This file was excluded because it contains continuous log2-transformed copy number values that require additional processing before they can be biologically interpreted. These values must be thresholded or converted into discrete categories, such as amplification or deletion, to be meaningful in downstream analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

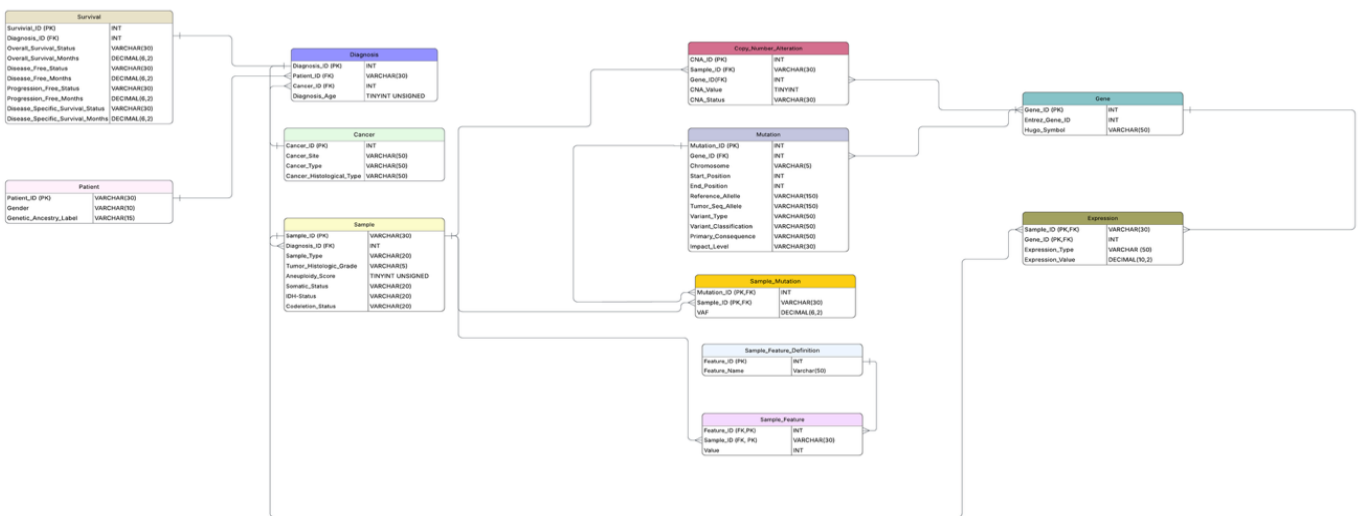
|                                                       |               |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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|                                                       |               |   | Since a processed CNA dataset with clearly defined gene-level states was already included, retaining this file would introduce redundancy and create potential inconsistency in how copy number alterations are represented across the database. In addition, using both representations could complicate integration with mutation and expression data. For these reasons, the log2 CNA file was excluded to maintain a single, consistent, and directly interpretable representation of copy number alterations.                                                                                                                                                                                                                                                                                                                                                              |
| data_methylation_hm27_hm450_merged.txt                | Full document | ✗ | The methylation dataset was excluded because it represents an epigenetic layer with very high dimensionality at the probe level, which requires specialized preprocessing, normalization, and analysis pipelines that differ from those used for mutation and CNA data. Including it would significantly increase complexity and shift the dataset beyond its defined scope, which is focused on gene-level genomic alterations rather than epigenetic regulation. In addition, probe-level methylation data does not integrate directly with the gene-centered structure of the database without substantial transformation.                                                                                                                                                                                                                                                   |
| data_mrna_seq_v2_rsem_zscores_ref_all_samples.txt     | Full document | ✗ | This file was excluded because it uses diploid samples as a reference for z-score normalization, introducing a different normalization framework. This creates inconsistency in how gene expression values are defined across datasets and complicates integration with other genomic layers. Maintaining multiple normalization strategies in the same database would increase the risk of analytical bias and confusion.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| data_mrna_seq_v2_rsem_zscores_ref_diploid_samples.txt | Full document | ✗ | This file was excluded because it uses diploid samples as a reference for z-score normalization, adding another layer of transformation that differs from both raw and globally normalized data. This creates inconsistency in how expression is represented across datasets and complicates integration. Maintaining multiple normalization schemes in the same database can lead to confusion and analytical bias.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| data_mrna_seq_v2_rsem.txt                             | Full document | ✓ | was included because it provides gene-level expression values in a consistent and scalable format that aligns with the structure of the mutation and CNA datasets. Each record is directly linked to a gene and a sample, which supports seamless integration across all data layers. Although the values are not z-score normalized, they are already processed through a standardized pipeline (RSEM), making them suitable for downstream normalization and analysis within a controlled framework. Keeping this file allows full flexibility in how expression data is handled, enabling consistent normalization alongside other datasets rather than relying on precomputed transformations that may introduce bias. This approach ensures that expression values remain interpretable, reproducible, and fully compatible with the gene-centered design of the database. |
| data_resource_patient.txt                             | Full document | ✗ | This was excluded because it contains only external resource links, such as imaging viewer URLs, which are not part of the genomic or molecular data used in this study. The variables included, such as Patient_ID, Resource_ID, and URL, serve as access points to external imaging systems rather than features that can be analyzed or integrated with mutation, CNA, or expression data. Including this file would not add any value to biological interpretation or downstream analysis.                                                                                                                                                                                                                                                                                                                                                                                  |
| data_rppa_zscores.txt                                 | Full document | ✗ | This file was excluded because it represents protein expression data in a z-score format, introducing an additional omics layer beyond the defined scope of this project. While the data is biologically meaningful, it follows a different structure and analytical framework, and integrating it would require separate preprocessing and interpretation pipelines that are not aligned with the current genomic-focused design.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| data_rppa.txt                                         | Full document | ✗ | This file was also excluded because it contains raw protein expression measurements that require normalization and specialized proteomics analysis. Including this dataset would introduce inconsistency in data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |



|                                      |                |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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|                                      |                |   | types and significantly increase complexity without directly supporting the mutation, CNA, and mRNA integration framework.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| data_timeline_status.txt             | Full document  | ✗ | This file was excluded because it captures longitudinal clinical information such as survival events, timing, and status changes over time. Although useful for detailed clinical studies, these time-based variables are not required for the current integration of genomic features and would add unnecessary complexity to the dataset.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| data_timeline_treatment.txt          | Full document  | ✗ | The data_timeline_treatment file was excluded because it includes detailed treatment histories such as chemotherapy, radiation, drug regimens, and dosages. While clinically important, these variables fall outside the scope of this project, which is focused on molecular alterations rather than treatment-response analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| data_timeline_sample_acquisition.txt | Full document  | ✗ | The data_timeline_sample_acquisition file was excluded because it contains metadata related to sample collection and processing, including procurement methods, tumor purity indicators, and handling details. These variables are not directly used in mutation, CNA, or expression analysis and do not contribute to the core genomic interpretation, so they were removed to maintain a clean and focused dataset.                                                                                                                                                                                                                                                                                                                                                                                      |
| data_sv.txt                          | Full document  | ✗ | This file was excluded because it contains structural variant information such as gene fusions and large chromosomal rearrangements, which require a different data model and analytical approach compared to point mutations and copy number alterations. Representing structural variants typically involves breakpoint-level information, partner genes, and complex event structures that do not fit within the simplified, gene-centered schema used in this database. Including this file would introduce additional complexity in both storage and interpretation, and would require separate processing and analysis pipelines. Since the scope of this project is focused on SNPs, small mutations, and gene-level CNA, structural variant data was excluded to maintain consistency and clarity. |
| Case_List.txt files                  | Full documents | ✗ | These files were excluded because they only define predefined sample groupings such as all samples or sequenced samples and do not contain any biological or molecular data. These files are primarily used for visualization tools and filtering within platforms like cBioPortal, rather than for analysis. Including them would not add any value to mutation, CNA, or expression analysis since the same information can already be derived directly from the dataset.                                                                                                                                                                                                                                                                                                                                 |
| Meta.txt files                       | Full document  | ✗ | These files were excluded because they contain dataset-level metadata such as file descriptions, data types, and formatting instructions rather than actual data. These files are useful for system configuration and data loading pipelines but do not contribute any variables or measurements that can be analyzed. Including them would introduce unnecessary information without enhancing the analytical or biological content of the database.                                                                                                                                                                                                                                                                                                                                                      |

3. **Diagrams:** A complete ERD showing all entities, attributes, and relationships, and the database schema normalized up to the Fifth Normal Form (5NF).





#### 4. Database Design Decisions

The schema is designed using a gene-centered, multi-omics integration framework to ensure consistent and unambiguous linkage across all data types. The core design principle is that every molecular observation must map to both a Sample and a Gene. This constraint enforces structural consistency and enables direct integration of mutation, copy number alteration (CNA), and gene expression data within a unified relational model.

The separation of Patient, Diagnosis, and Sample reflects the underlying clinical hierarchy. A patient may have multiple diagnoses, and each diagnosis may generate one or more samples. Although the current dataset contains a single sample per patient, this structure is intentionally preserved to ensure scalability and compatibility with real-world clinical scenarios involving longitudinal sampling, disease progression, or multiple biopsies.

The schema adopts a modular design with dedicated entities for each biological layer. The Gene table is defined as an independent reference entity to enforce consistent gene identification across all datasets using standardized identifiers such as Hugo Symbol and Entrez Gene ID. Molecular data types, including Mutation, Copy\_Number\_Alteration, and Expression, are modeled as separate entities to preserve their structural integrity while enabling integration through shared foreign keys.

Many-to-many relationships are explicitly resolved using bridge tables. The Sample\_Mutation table is a key example, linking samples and mutations while storing Variant Allele Frequency (VAF) as an attribute of the relationship. This design eliminates duplication, maintains normalization, and supports efficient querying at both mutation-level and gene-level resolutions.

Additional modular components are introduced to improve flexibility and extensibility. The Survival table isolates multiple survival endpoints (OS, DSS, DFS, PFS), preventing redundancy and ensuring consistent representation of clinical outcomes. The Sample\_Feature and Feature\_Definition tables provide a flexible mechanism for extending sample-level attributes without modifying the core schema, allowing future incorporation of additional derived or experimental features.

Overall, the schema is designed to:

- Enforce strict gene-sample mapping across all molecular data
- Eliminate redundancy through normalization and decomposition
- Support scalable integration of multi-omics datasets
- Maintain clear, consistent, and biologically meaningful relationships
- Enable efficient querying and future extensibility

#### 5. Normal Form and Deviations

The database schema is normalized up to the Fifth Normal Form (5NF) to minimize redundancy and enforce strong data integrity across all integrated datasets.

First Normal Form (1NF) is satisfied by ensuring that all attributes are atomic and that no repeating groups or multi-valued fields exist. Each table contains only indivisible values, which simplifies querying and prevents ambiguity in data representation.

Second Normal Form (2NF) is achieved by eliminating partial dependencies. All non-key attributes depend on the full primary key. This is particularly important in tables with composite keys, such as Sample\_Mutation, where attributes like Variant Allele Frequency (VAF) depend on the combination of both Sample\_ID and Mutation\_ID rather than on either key individually.

Third Normal Form (3NF) is enforced by removing transitive dependencies. Attributes are stored only in the tables where they logically belong. For example, gene-related attributes are stored exclusively in the Gene table and are not duplicated in Mutation, Copy\_Number\_Alteration, or Expression tables. This prevents inconsistency and ensures a single source of truth for each entity.



Higher normalization levels, including Boyce–Codd Normal Form (BCNF), Fourth Normal Form (4NF), and Fifth Normal Form (5NF), are addressed through systematic decomposition of relationships and explicit handling of many-to-many associations. Bridge tables, such as Sample\_Mutation, are used to fully resolve these relationships, ensuring that all dependencies are properly represented without introducing redundancy or hidden constraints.

No major deviations from normalization are introduced. However, controlled and intentional design choices are applied where necessary for performance and usability. The primary example is the storage of derived attributes such as Variant Allele Frequency (VAF). Although VAF can be computed from read counts, it is stored directly to improve query efficiency, reduce repeated computation, and simplify downstream analysis while maintaining consistency with the underlying data.

This normalization strategy ensures:

- Elimination of redundancy across all entities
- Preservation of data integrity and consistency
- Clear separation of concerns between data layers
- Scalability for future data expansion
- Reliable integration of multi-omics datasets within a unified schema

## 6. Data Dictionary

The database is structured around well-defined entities, where each table represents a distinct biological or analytical concept. The design enforces consistent naming conventions, clear attribute definitions, and strict relationships between entities to support reliable multi-omics integration and querying.

### Patient

- Patient\_ID (VARCHAR, PK): Unique identifier for each patient
- Gender (VARCHAR): Biological sex of the patient
- Genetic\_Ancestry\_Label (VARCHAR): Population-level ancestry classification used for genomic context

### Diagnosis

- Diagnosis\_ID (INT, PK): Unique identifier for each diagnosis
- Patient\_ID (VARCHAR, FK): Links diagnosis to patient (i.e., References Patient.Patient\_ID)
- Cancer\_ID (INT, FK): Links diagnosis to cancer type (i.e., References Cancer.Cancer\_ID)
- Diagnosis\_Age (TINYINT): Age of the patient at diagnosis

### Cancer

- Cancer\_ID (INT, PK): Unique cancer identifier
- Cancer\_Site (VARCHAR): Anatomical site (e.g., Brain)
- Cancer\_Type (VARCHAR): Cancer category (e.g., Low-Grade Glioma)
- Cancer\_Histological\_Type (VARCHAR): Histological classification (e.g., astrocytomas, oligodendrogliomas, and oligoastrocytomas)

### Sample

- Sample\_ID (VARCHAR, PK): Unique identifier for each sample
- Diagnosis\_ID (INT, FK): Links sample to diagnosis (i.e., References Diagnosis.Diagnosis\_ID)
- Tumor\_Histologic\_Grade (VARCHAR): Tumor grade classification
- Aneuploidy\_Score (INT): Measure of chromosomal instability
- Somatic\_Status (VARCHAR): Indicates whether alterations are somatic
- IDH\_Status (VARCHAR): IDH mutation status (e.g., Mutant, Wildtype)
- Codeletion\_Status (VARCHAR): 1p/19q codeletion status (e.g., Codeleted, Noncodeleted)

### Gene

- Gene\_ID (INT, PK): Internal unique gene identifier
- Hugo\_Symbol (VARCHAR): Standard gene symbol
- Entrez\_Gene\_ID (INT): Standard gene identifier

### Mutation

- Mutation\_ID (INT, PK): Unique mutation identifier
- Gene\_ID (INT, FK): Links mutation to gene (i.e., References Gene.Gene\_ID)
- Chromosome (VARCHAR): Chromosomal location
- Start\_Position (INT): Genomic start position
- End\_Position (INT): Genomic end position
- Reference\_Allele (VARCHAR): Reference nucleotide sequence
- Tumor\_Seq\_Allele (VARCHAR): Observed (i.e., Tumor) nucleotide sequence
- Variant\_Classification (VARCHAR): Functional mutation class
- Variant\_Type (VARCHAR): Type of mutation (e.g., SNP, insertion, deletion)
- Primary\_Consequence (VARCHAR): Primary functional effect of the mutation
- Impact\_Level (VARCHAR): Predicted level of biological impact (i.e., Low, Moderate, High)

### Sample\_Mutation\*\*

- Sample\_ID (VARCHAR, FK): Links to sample (i.e, References Sample.Sample\_ID)
- Mutation\_ID (INT, FK): Links to mutation (i.e, References Mutation.Mutation\_ID)
- VAF (DECIMAL): Variant Allele Frequency
- Composite Primary Key: (Sample\_ID, Mutation\_ID)

**\*\*This table resolves the many-to-many relationship between samples and mutations, storing mutation prevalence within each sample.**

#### Copy\_Number\_Alteration

- CNA\_ID (INT, PK): Unique CNA record
- Sample\_ID (VARCHAR, FK): Links to sample (i.e, References Sample.Sample\_ID)
- Gene\_ID (INT, FK): Links to gene (i.e, References Gene.Gene\_ID)
- CNA\_Value (INT): Numeric copy number value
- CNA\_Status (VARCHAR): Interpreted categorical state (e.g, gain, loss, amplification, etc.)

#### Expression\*\*

- Sample\_ID (VARCHAR, FK): Links to sample (i.e, References Sample.Sample\_ID)
- Gene\_ID (INT, FK): Links to gene (i.e, References Gene.Gene\_ID)
- Expression\_Type (VARCHAR): Type of expression measurement
- Expression\_Value (DECIMAL): Expression level (i.e, Gene expression level)
- Composite Primary Key: (Sample\_ID, Gene\_ID)

**\*\* This table captures gene-level expression values and ensures one record per gene-sample pair.**

#### Survival\*\*

- Survival\_ID (INT, PK): Unique survival record
- Diagnosis\_ID (INT, FK): Links to diagnosis (i.e, References Diagnosis.Diagnosis\_ID)
- Overall\_Survival\_Status (VARCHAR): Patient survival status
- Overall\_Survival\_Months (DECIMAL): Survival time in months
- Disease\_Specific\_Survival\_Status (VARCHAR): Disease-specific survival status
- Disease\_Specific\_Survival\_Months (DECIMAL): Disease-specific survival time
- Disease\_Free\_Status (VARCHAR): Recurrence status
- Disease\_Free\_Months (DECIMAL): Time to recurrence
- Progression\_Free\_Status (VARCHAR): Progression status
- Progression\_Free\_Months (DECIMAL): Time to progression

**\*\* This table separates survival outcomes to avoid redundancy and maintain clarity across endpoints.**

#### Feature\_Definition

- Feature\_ID (INT, PK): Unique feature identifier
- Feature\_Name (VARCHAR): Name of the feature

#### Sample\_Feature\*\*

- Feature\_ID (INT, FK): Links to feature definition (i.e, References Feature\_Definition.Feature\_ID)
- Sample\_ID (VARCHAR, FK): Links to sample (i.e, References Sample.Sample\_ID)
- Value (INT): Feature value
- Composite Primary Key: (Feature\_ID, Sample\_ID)

**\*\* This structure supports flexible extension of sample-level features without modifying the core schema.**

This data dictionary enforces a consistent and normalized structure where each entity has a clearly defined role. Relationships are explicit, redundancy is minimized, and all data layers remain directly linkable through shared keys. This design ensures reliable querying, scalability, and compatibility with multi-omics analysis workflows.

## 7. Script Map & Data Cleaning Decisions

**Table 2. Reproducible Python Pipeline (executed in the order outlined) for Gene-Centered Multi-Omics Data Cleaning (with explicit transformation decisions outlining exactly what changed), Harmonization, and Integration**

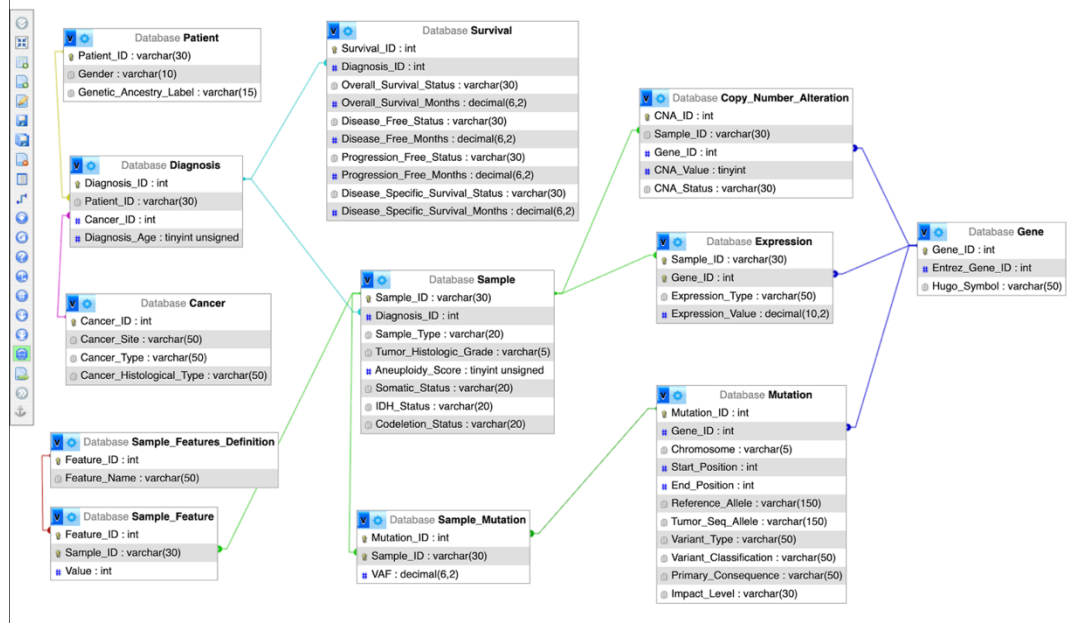
| Stage | Script                      | Data Cleaning and Transformation Description                                                                                                                                                                                                                                                                                                                                                                                                 | Input Dataset           | Output Dataset       |
|-------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------|
|       | 01_clean_clinical_sample.py | Cleans and standardizes clinical and sample data into a structured format ready for analysis and database integration. It renames variables into consistent, database-aligned names, converts encoded survival variables into clear clinical labels, and rounds time-related fields to keep everything consistent. It also standardizes cancer type and site naming, then extracts IDH status and 1p/19q codeletion status directly from the | Clinical&Sample(1).xlsx | Clinical&Sample.xlsx |

|          |                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           |                     |
|----------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------|
| Cleaning |                        | <p>subtype information. Finally, it filters and orders the dataset into a fixed schema to improve visualization and outputs a clean, standardized, and analysis-ready file. All variable names are unified to match the database structure and ensure smooth downstream use.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                           |                     |
|          |                        | <p><b>What exactly changed?</b></p> <ul style="list-style-type: none"> <li>- Standardized all column names to a consistent database-friendly format (e.g., PATIENT_ID → Patient_ID, SAMPLE_ID → Sample_ID, SEX → Gender, AGE → Diagnosis_Age).</li> <li>- Renamed all survival variables to clear and descriptive clinical terms (e.g., OS_STATUS → Overall_Survival_Status, OS_MONTHS → Overall_Survival_Months, etc.).</li> <li>- Converted binary survival encodings (0/1) into explicit clinical labels based on cBioPortal definitions to avoid ambiguity (e.g., Living/Deceased, Disease Free/Recurred).</li> <li>- Rounded all survival time variables (OS_MONTHS, DSS_MONTHS, DFS_MONTHS, PFS_MONTHS) to two decimal places to ensure numerical consistency.</li> <li>- Standardized cancer-related variables for clarity (e.g., CANCER_TYPE_ACRONYM → Cancer_Type, expanded “LGG” to “Low-Grade Glioma”; TUMOR_TISSUE_SITE → Cancer_Site, replaced “CNS” with “Brain”).</li> <li>- Renamed CANCER_TYPE_DETAILED to Cancer_Histological_Type to accurately reflect histological classification (e.g., astrocytomas, oligodendrogliomas, and oligoastrocytomas).</li> <li>- Split the Subtype variable into two biologically meaningful features: IDH_Status and Codeletion_Status, and removed redundant “LGG” prefix.</li> <li>- Standardized additional variables without changing their values (e.g., GRADE → Tumor_Histologic_Grade, ANEUPLOIDY_SCORE → Aneuploidy_Score, SAMPLE_TYPE → Sample_Type, SOMATIC_STATUS → Somatic_Status).</li> <li>- Renamed hypoxia-related variables to a consistent format (BUFFA_HYPOXIA_SCORE, WINTER_HYPOXIA_SCORE, RAGNUM_HYPOXIA_SCORE → Buffa_Hypoxia_Score... etc).</li> <li>- Trimmed whitespace from column names and values to eliminate formatting inconsistencies.</li> <li>- Verified dataset uniqueness using Patient_ID and Sample_ID; no exact duplicate rows were found, so no records were removed.</li> </ul> |                           |                     |
|          | 02_clean_cna.py        | <p>Transforms CNA data from wide to long format, where each row represents a gene–sample pair. It standardizes gene identifiers, converts copy number values into clear biological labels (i.e, status), handles missing values explicitly, and removes duplicates while preserving the original structure. The output is a clean, analysis-ready table optimized for integration with mutation, expression, and clinical datasets. All variable names are unified to match the database structure.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | data_cna.txt              | cna_long.csv        |
|          |                        | <p><b>What exactly changed?</b></p> <ul style="list-style-type: none"> <li>- Converted the CNA dataset from wide matrix format (samples as columns) into long format where each row represents a unique gene-sample pair (Hugo_Symbol, Entrez_Gene_Id, Sample_ID).</li> <li>- Standardized gene identifiers by enforcing correct data types: Entrez_Gene_Id converted to numeric (nullable integer) and Hugo_Symbol enforced as string.</li> <li>- Renamed the CNA value column to CNA_Value during transformation to clearly represent numeric copy number states.</li> <li>- Converted all CNA values to numeric format, with invalid or non-numeric entries coerced to missing values (NaN).</li> <li>- Mapped numeric CNA values to biologically meaningful categorical labels stored in a new column CNA_Status: -2 → deep_loss , -1 → loss, 0 → neutral , 1 → gain , 2 → amplification</li> <li>- Removed exact duplicate rows to eliminate redundancy while preserving the original gene-sample relationships.</li> <li>- Verified dataset size before and after duplicate removal to ensure transparency and data integrity.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           |                     |
|          | 03_clean_expression.py | <p>Converts gene expression data from wide to long format, where each row represents a gene–sample pair. It standardizes gene identifiers, cleans and rounds expression values, removes zero-expression and duplicate rows, and outputs a structured dataset ready for analysis and integration. All variable names are unified to match the database structure and support efficient multi-omics workflows.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | data_mrna_seq_v2_rsem.txt | long_expression.csv |
|          |                        | <p><b>What exactly changed?</b></p> <ul style="list-style-type: none"> <li>- Converted the RNA-seq dataset from wide matrix format (samples as columns) into long format where each row represents a unique gene–sample pair (Hugo_Symbol, Entrez_Gene_Id, Sample_ID).</li> <li>- Standardized gene identifiers by enforcing correct data types: Entrez_Gene_Id converted to numeric (nullable integer) and Hugo_Symbol enforced as string.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                           |                     |

|                             |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                       |                                                                      |
|-----------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|                             |                            | <ul style="list-style-type: none"> <li>- Renamed the expression value column to Expression to clearly represent mRNA expression levels.</li> <li>- Converted all expression values to numeric format, with invalid entries coerced to missing values.</li> <li>- Rounded expression values to two decimal places to ensure consistency and reduce file size without affecting interpretability.</li> <li>- Removed rows with zero expression values to eliminate non-informative data and improve computational efficiency.</li> <li>- Removed exact duplicate rows to ensure data integrity and avoid redundancy in gene-sample relationships.</li> <li>- Verified dataset size before and after duplicate removal to maintain transparency in data cleaning.</li> </ul>                                                                                                                                                                                                     |                                                                                                       |                                                                      |
|                             | 04_clean_mutations.py      | Processes mutation data by renaming the consequence field to represent the primary consequence, then selects the most severe variant for each mutation based on VEP severity ranking. It standardizes mutation annotations into a consistent format, calculates variant allele frequency from read counts, and rounds it to two decimal places to ensure consistency. Finally, it outputs a clean, structured dataset ready for analysis and database integration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Mutation(1).xlsx                                                                                      | Mutation.xlsx                                                        |
|                             |                            | <p><b>What exactly changed?</b></p> <ul style="list-style-type: none"> <li>- Renamed key columns to align with the database schema (Consequence → Primary_Consequence, IMPACT → Impact_Level).</li> <li>- Parsed multiple consequence annotations per mutation and selected only the most severe consequence using official VEP severity ranking.</li> <li>- Standardized mutation annotation fields (Primary_Consequence, Variant_Classification, Variant_Type, Impact_Level) by normalizing text format (lowercase → capitalized) to ensure consistency.</li> <li>- Converted read count fields (t_ref_count, t_alt_count) to numeric format, with invalid values coerced to missing values.</li> <li>- Calculated Variant Allele Frequency (VAF) using the formula:<br/> <math display="block">VAF = t\_alt\_count / (t\_ref\_count + t\_alt\_count)</math> </li> <li>- Rounded VAF values to two decimal places to ensure consistency and improve readability.</li> </ul> |                                                                                                       |                                                                      |
| Sample & Gene Harmonization | 05_validator.py            | Validates input datasets by enforcing the presence of key fields such as Sample_ID and Hugo_Symbol, checking for missing or invalid identifiers, standardizing gene symbols, removing invalid or empty entries, and verifying that the HGNC reference file contains the required mapping columns. This ensures that only clean, consistent, and reliable data enters the pipeline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Clinical&Sample.xlsx<br>long_expression.csv<br>Mutation.xlsx<br>cna_long.csv<br>hgnc_complete_set.txt |                                                                      |
|                             | 06_sample_harmonization.py | Harmonizes datasets at the sample level by cleaning and standardizing Sample_ID values, validating required fields, identifying the intersection of samples shared across expression, CNA, mutation, and clinical datasets, and filtering each dataset to retain only these common samples. This ensures that all retained samples have complete multi-omics data and produces matched datasets ready for downstream integration and analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Clinical&Sample.xlsx<br>long_expression.csv<br>Mutation.xlsx<br>cna_long.csv                          | expr_step1.csv<br>cna_step1.csv<br>mut_step1.xlsx<br>clin_step1.xlsx |
|                             | 07_gene_harmonization.py   | Harmonizes datasets at the gene level by standardizing Hugo_Symbol values, identifying the intersection of genes shared across expression, CNA, and mutation datasets, and filtering each dataset to retain only these common genes. This ensures consistent gene representation across all modalities and produces matched datasets ready for reliable multi-omics integration and analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | expr_step1.csv<br>cna_step1.csv<br>mut_step1.xlsx                                                     | expr_step2.csv<br>cna_step2.csv<br>mut_step2.xlsx                    |
|                             | 08_hgnc_mapping.py         | Standardizes gene identifiers using HGNC mapping by aligning Hugo_Symbol values to Entrez Gene IDs, incorporating official symbols, aliases, and previous gene names to resolve inconsistencies, and filtering out unmapped entries. This ensures unified gene identifiers across all datasets and produces matched outputs ready for reliable multi-omics integration and analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | expr_step3.csv<br>cna_step3.csv<br>mut_step3.xlsx                                                     | expr_step3.csv<br>cna_step3.csv<br>mut_step3.xlsx                    |
|                             | 09_finalize.py             | Performs final harmonization across all datasets by enforcing both sample-level and gene-level alignment. It retains only samples shared across expression, CNA, mutation, and clinical datasets, and ensures that                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | expr_step3.csv<br>cna_step3.csv<br>mut_step3.xlsx<br>clin_step1.xlsx                                  | expr_FINAL.csv<br>cna_FINAL.csv<br>mut_FINAL.xlsx<br>clin_FINAL.xlsx |

|                           |                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                      |                        |
|---------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------|
| <div></div>               |                                           | expression, CNA, and mutation data contain only common genes. This produces fully matched and consistent datasets ready for reliable multi-omics integration, database ingestion, and downstream analysis.                                                                                                                                                                                                                                                                                                                                               |                                                                      |                        |
|                           | 10_report.py                              | Generates a final validation and proof report by computing key dataset metrics, including counts of patients, samples, genes, mutations, and omics rows. It deduplicates mutations and sample-mutation pairs to match database population logic, summarizes cancer definitions and clinical features, and evaluates overall dataset completeness. This report serves both as a data integrity check and as a formal proof that the datasets are consistent, accurately structured, and ready for database ingestion and downstream multi-omics analysis. | expr_FINAL.csv<br>cna_FINAL.csv<br>mut_FINAL.xlsx<br>clin_FINAL.xlsx | final_proof_report.txt |
|                           | run_pipeline.py                           | Orchestrates the full pipeline by executing all steps in sequence, including sample-level harmonization, gene-level harmonization, HGNC-based gene identifier mapping, final dataset alignment, and validation report generation. This ensures a fully automated, consistent, and reproducible multi-omics integration workflow ready for database ingestion and downstream analysis..                                                                                                                                                                   |                                                                      |                        |
| Database Population       | SQL_generation_for_populating_the_data.py | Builds a complete SQL population script by transforming the final datasets into structured INSERT statements, standardizing values, mapping relationships between patients, samples, genes, and mutations, and deduplicating records to match database constraints. The output is a normalized, ready-to-run script for MySQL database population.                                                                                                                                                                                                       | clin_FINAL.xlsx<br>mut_FINAL.xlsx<br>cna_FINAL.csv<br>expr_FINAL.csv | FINAL_POPULATE.sql     |
| Graph Database Population | Neo4j.py                                  | Generates Neo4j Cypher queries from a CSV file exported from SQL-based exploratory queries. It standardizes sample and gene data, assigns IDH and 1p/19q codeletion attributes to sample nodes, creates gene nodes, and defines relationships between samples and genes. The output is a clean, ready-to-run script for graph database import and network-based analysis. Neo4j.py                                                                                                                                                                       | For Neo4j.csv                                                        | neo4j_import.cypher    |

Figure 2: Implemented Database Schema on the Virtual Machine (Entity-Relationship View)



## Database Population on the Virtual Machine

| Table                                               | Action | Rows      | Type   | Collation          | Size      | Overhead |
|-----------------------------------------------------|--------|-----------|--------|--------------------|-----------|----------|
| <input type="checkbox"/> Cancer                     |        | 3         | InnoDB | utf8mb4_0900_ai_ci | 16.0 KiB  | -        |
| <input type="checkbox"/> Copy_Number_Alteration     |        | 1,150,724 | InnoDB | utf8mb4_0900_ai_ci | 48.0 KiB  | -        |
| <input type="checkbox"/> Diagnosis                  |        | 499       | InnoDB | utf8mb4_0900_ai_ci | 80.0 KiB  | -        |
| <input type="checkbox"/> Expression                 |        | 5,514,987 | InnoDB | utf8mb4_0900_ai_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> Gene                       |        | 12,311    | InnoDB | utf8mb4_0900_ai_ci | 464.0 KiB | -        |
| <input type="checkbox"/> Mutation                   |        | 33,653    | InnoDB | utf8mb4_0900_ai_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> Patient                    |        | 499       | InnoDB | utf8mb4_0900_ai_ci | 16.0 KiB  | -        |
| <input type="checkbox"/> Sample                     |        | 499       | InnoDB | utf8mb4_0900_ai_ci | 80.0 KiB  | -        |
| <input type="checkbox"/> Sample_Feature             |        | 1,497     | InnoDB | utf8mb4_0900_ai_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> Sample_Features_Definition |        | 3         | InnoDB | utf8mb4_0900_ai_ci | 16.0 KiB  | -        |
| <input type="checkbox"/> Sample_Mutation            |        | 34,282    | InnoDB | utf8mb4_0900_ai_ci | 32.0 KiB  | -        |
| <input type="checkbox"/> Survival                   |        | 499       | InnoDB | utf8mb4_0900_ai_ci | 64.0 KiB  | -        |
| 12 table(s)                                         | Sum    | 6,749,456 | InnoDB | utf8mb4_0900_ai_ci | 912 KiB   | 0 B      |

☐ Check all    With selected:

final\_proof\_report.txt

Patients: 499  
Cancers: 3  
Diagnoses: 499  
Survival rows: 499  
Samples: 499  
Genes: 12311  
Mutations: 33653  
Sample\_Mutation rows: 34282  
CNA rows: 1150724  
Expression rows: 5514987  
Feature definitions: 3  
Sample\_Feature rows: 1497

This figure displays the successfully populated database on the virtual machine, alongside the validation report generated in Step 5 of the harmonization stage outlined earlier. The database tables reflect the inserted records across all entities. At the same time, the validation report confirms data integrity through consistent counts of patients (499), samples (499), genes (12,311), mutations (33,653), and multi-omics records, including expression (5,514,987 rows) and CNA data (1,150,724 rows). The matching statistics between the database and the generated report provide formal proof of correct data population and structural consistency.

## 8. General SQL Queries with Rationale and Resulting Outputs

Please note that this is often not the complete output. It only reflects how the results are displayed.

### 1. SQL:

```
1 SELECT Sample.Sample_ID
2 FROM Patient
3 JOIN Diagnosis ON Patient.Patient_ID = Diagnosis.Patient_ID
4 JOIN Sample ON Diagnosis.Diagnosis_ID = Sample.Diagnosis_ID
5 WHERE Patient.Patient_ID = 'TCGA-CS-4938';
```

#### Rationale:

This query uses the Patient → Diagnosis → Sample relationship to accurately retrieve all samples associated with a specific patient. It follows the correct biological hierarchy, where samples are linked to diagnoses rather than directly to patients. This ensures data integrity and avoids incorrect mappings. Although the current dataset contains one sample per patient, this structure maintains scalability for future scenarios with multiple samples per diagnosis. It also acts as a validation step to confirm that the relationships between tables are correctly implemented and consistent.

#### Output:

**Sample\_ID**  
TCGA-CS-4938-01

### 2. SQL:

```
1 SELECT
2     Impact_Level,
3     COUNT(*) AS Count
4 FROM Sample_Mutation
5 JOIN Mutation
6     ON Sample_Mutation.Mutation_ID = Mutation.Mutation_ID
7 WHERE Sample_Mutation.Sample_ID = 'TCGA-CS-4938-01'
8 GROUP BY Impact_Level;
```

#### Rationale:

This query uses the Sample\_Mutation bridge table to correctly resolve the many-to-many relationship between samples and mutations, then joins with the Mutation table to access functional impact annotations. It aggregates mutations by Impact\_Level to produce a clear summary of mutation severity within a specific sample. This approach ensures accurate counting without duplication and enables consistent



comparison across samples. It also supports scalability, as the same structure can handle large mutation datasets and more complex annotation layers without modification.

Output:

| Impact_Level | Count |
|--------------|-------|
| Low          | 5     |
| Moderate     | 12    |
| High         | 2     |

3. SQL:

```
1 SELECT DISTINCT Gene.Hugo_Symbol
2 FROM Sample_Mutation
3 JOIN Mutation ON Sample_Mutation.Mutation_ID = Mutation.Mutation_ID
4 JOIN Gene ON Mutation.Gene_ID = Gene.Gene_ID
5 WHERE Sample_Mutation.Sample_ID = 'TCGA-CS-4938-01';
```

Rationale:

This query traverses the Sample\_Mutation → Mutation → Gene relationship to map all mutations in a sample to their corresponding genes. It uses DISTINCT to ensure each gene is reported once, avoiding duplication from multiple mutations within the same gene. This structure enables accurate transformation from mutation-level data to gene-level interpretation, which is more biologically meaningful.

Output:

Hugo\_Symbol

CELF6

GPR114

TP53

KLK10

C3

ANXA9

TTLL7

TPO

IDH1

LEPREL1

ENPEP

ADAD1

TKTL2

FGF5

ADAM22

ACTRT1

ZAN

FAM71F2

CDC47L

4. SQL:

```
1 SELECT
2     Gene.Hugo_Symbol,
3     Copy_Number_Alteration.CNA_Value,
4     Copy_Number_Alteration.CNA_Status
5 FROM Copy_Number_Alteration
6 JOIN Gene
7     ON Copy_Number_Alteration.Gene_ID = Gene.Gene_ID
8 WHERE Copy_Number_Alteration.Sample_ID = 'TCGA-CS-4938-01';
```

Rationale:

This query joins Copy\_Number\_Alteration with Gene to map gene-level copy number changes within a specific sample. It retrieves both the numeric CNA\_Value and its categorical interpretation (CNA\_Status), ensuring clear and consistent representation of genomic alterations. This structure preserves accurate gene-level mapping, avoids redundancy, and supports scalable analysis across samples. It also enables identification of broader chromosomal patterns, as multiple genes sharing the same CNA status can indicate regional amplification or deletion events rather than isolated gene-specific changes.

Output:

| Hugo_Symbol | CNA_Value | CNA_Status |
|-------------|-----------|------------|
| ABCB7       | 1         | gain       |
| ABCB8       | 1         | gain       |
| ABCD1       | 1         | gain       |
| ABCD4       | -1        | loss       |
| ABCF2       | 1         | gain       |
| ACE2        | -1        | loss       |
| ACHE        | 1         | gain       |
| ACN9        | 1         | gain       |
| ACOT1       | -1        | loss       |
| ACOT4       | -1        | loss       |

5. SQL:

```
1 SELECT Survival.Overall_Survival_Status, Survival.Overall_Survival_Months
2 FROM Patient
3 JOIN Diagnosis ON Patient.Patient_ID = Diagnosis.Patient_ID
4 JOIN Survival ON Diagnosis.Diagnosis_ID = Survival.Diagnosis_ID
5 WHERE Patient.Patient_ID = 'TCGA-CS-4938';
```

Rationale:

This query follows the Patient → Diagnosis → Survival relationship to accurately retrieve survival outcomes for a specific patient. It ensures that survival data is correctly linked to the corresponding diagnosis, avoiding misalignment between clinical and patient records. This structure maintains data integrity and supports scalability for cases with multiple diagnoses per patient. It also enables consistent integration of clinical outcomes with molecular data for downstream analysis.

Output:

| Overall_Survival_Status | Overall_Survival_Months |
|-------------------------|-------------------------|
| Living                  | 117.50                  |

6. SQL:

```
1 SELECT Sample_Mutation.Sample_ID, Sample_Mutation.VAF
2 FROM Sample_Mutation
3 WHERE Sample_Mutation.VAF > 0.5;
```

Rationale:

This query directly filters the Sample\_Mutation table using a VAF threshold to identify mutations with high allele frequency within samples. It avoids unnecessary joins since VAF is already stored at the sample–mutation level, ensuring efficient execution. This approach preserves accurate mutation-level representation while enabling scalable screening across large datasets. It also provides a consistent method to prioritize dominant variants for downstream analysis and comparison across samples.

Output:

| Sample_ID       | VAF  |
|-----------------|------|
| TCGA-CS-4938-01 | 0.67 |
| TCGA-CS-4938-01 | 0.81 |
| TCGA-DB-A4XG-01 | 0.53 |
| TCGA-DB-A64V-01 | 0.52 |

7. SQL:

```
1 SELECT Sample_Mutation.Sample_ID, COUNT(*) AS mutation_count
2 FROM Sample_Mutation
3 GROUP BY Sample_Mutation.Sample_ID
4 ORDER BY mutation_count DESC;
```

Rationale:

This query aggregates mutations at the sample level using GROUP BY to compute the total mutation count per sample. It operates directly on the Sample\_Mutation table, ensuring accurate counting without duplication. The ORDER BY clause ranks samples from highest to

lowest mutation burden, enabling quick identification of highly mutated or outlier samples. This structure is efficient, scalable, and supports consistent comparison across all samples for downstream analysis.

**Output:**

| Sample_ID       | mutation_count ▾ 1 |
|-----------------|--------------------|
| TCGA-DU-6392-01 | 14002              |
| TCGA-HT-8564-01 | 613                |
| TCGA-DU-7294-01 | 537                |
| TCGA-DU-7298-01 | 358                |
| TCGA-DU-A7T6-01 | 126                |
| TCGA-HW-7490-01 | 119                |
| TCGA-DU-7010-01 | 107                |
| TCGA-P5-A5EV-01 | 98                 |
| TCGA-HT-7688-01 | 96                 |
| TCGA-HT-7860-01 | 93                 |
| TCGA-DB-A64L-01 | 92                 |
| TCGA-DB-A64X-01 | 90                 |

**8. SQL:**

```
1 SELECT Feature_Name, Value
2 FROM Sample_Feature, Sample_Features_Definition
3 WHERE Sample_Feature.Feature_ID = Sample_Features_Definition.Feature_ID
4 AND Sample_ID = 'TCGA-CS-4938-01';
```

**Rationale:**

This query connects the sample features with their definitions so you can actually understand what each value means. Instead of just seeing feature IDs, you get clear feature names along with their values for the selected sample. This makes the data easier to read and interpret, especially when comparing features across samples. It also keeps the structure clean and flexible, so you can add more features later without changing how the query works.

**Output:**

| Feature_Name         | Value |
|----------------------|-------|
| Bufa_Hypoxia_Score   | -45   |
| Winter_Hypoxia_Score | -48   |
| Ragnum_Hypoxia_Score | -18   |

**9. SQL:**

```
1 SELECT DISTINCT Sample_ID
2 FROM Copy_Number_Alteration
3 WHERE CNA_Status = 'amplification';
```

**Rationale:**

This query filters the copy number data to find samples that show gene amplification. It uses DISTINCT to make sure each sample appears only once, even if multiple genes are amplified in the same sample. This keeps the output clean and avoids repetition. The query is simple and efficient because it works directly on the CNA table without extra joins, and it can easily scale to larger datasets when you want to screen for samples with specific genomic events.

**Output:**

| Sample_ID       |
|-----------------|
| TCGA-CS-4941-01 |
| TCGA-CS-4942-01 |
| TCGA-CS-4943-01 |
| TCGA-CS-4944-01 |
| TCGA-CS-5390-01 |
| TCGA-CS-5393-01 |

## 10. SQL:

```

1 SELECT
2     Patient.Patient_ID,
3     Sample.Sample_ID,
4     Gene.Hugo_Symbol,
5     Mutation.Primary_Consequence,
6     Sample_Mutation.VAF
7 FROM Patient
8 JOIN Diagnosis ON Patient.Patient_ID = Diagnosis.Patient_ID
9 JOIN Sample ON Diagnosis.Diagnosis_ID = Sample.Diagnosis_ID
10 JOIN Sample_Mutation ON Sample.Sample_ID = Sample_Mutation.Sample_ID
11 JOIN Mutation ON Sample_Mutation.Mutation_ID = Mutation.Mutation_ID
12 JOIN Gene ON Mutation.Gene_ID = Gene.Gene_ID;

```

## Rationale:

This query brings everything together in one place. It connects patient data to their samples, then links those samples to mutations and finally to the affected genes. This way, you move step by step from the patient level down to the exact genetic changes. It keeps the relationships correct, so each mutation is tied to the right sample and patient. It also avoids mixing data from unrelated records. The result is a clear, detailed view that shows which genes are mutated in each sample and how strong those mutations are using VAF. This structure makes it easy to explore the data, compare patients, and connect genetic changes with clinical context without rewriting multiple queries.

## Output:

| Patient_ID   | Sample_ID       | Hugo_Symbol | Primary_Consequence     | VAF  |
|--------------|-----------------|-------------|-------------------------|------|
| TCGA-CS-4938 | TCGA-CS-4938-01 | CELF6       | Synonymous_variant      | 0.67 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | GPR114      | Missense_variant        | 0.11 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | TP53        | Missense_variant        | 0.81 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | KLK10       | Synonymous_variant      | 0.33 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | C3          | Stop_gained             | 0.15 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | ANXA9       | Splice_acceptor_variant | 0.07 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | TTLL7       | Synonymous_variant      | 0.48 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | TPO         | Missense_variant        | 0.16 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | IDH1        | Missense_variant        | 0.36 |
| TCGA-CS-4938 | TCGA-CS-4938-01 | LEPREL1     | Missense_variant        | 0.22 |

## 11. SQL:

```

1 SELECT Gene.Hugo_Symbol, COUNT(*) AS mutation_frequency
2 FROM Sample_Mutation
3 JOIN Mutation ON Sample_Mutation.Mutation_ID = Mutation.Mutation_ID
4 JOIN Gene ON Mutation.Gene_ID = Gene.Gene_ID
5 GROUP BY Gene.Hugo_Symbol
6 ORDER BY mutation_frequency DESC
7 LIMIT 10;

```

## Rationale:

This query counts how often each gene is mutated across all samples, then ranks them from most frequent to least. It links mutations to their genes and uses GROUP BY to combine all mutations belonging to the same gene. The ORDER BY and LIMIT help you focus only on the top 10 genes.

This keeps the result clear and avoids duplication, even when a gene has many mutations across different samples.

## Output:

| Hugo_Symbol | mutation_frequency |
|-------------|--------------------|
| TP53        | 291                |
| IDH1        | 286                |
| ATRX        | 214                |
| TTN         | 159                |
| CIC         | 126                |
| MUC16       | 66                 |
| FUBP1       | 51                 |
| NOTCH1      | 49                 |
| NF1         | 46                 |
| EGFR        | 43                 |

## 12. SQL:

```

1 SELECT
2     Sample_Mutation.Sample_ID,
3     AVG(Sample_Mutation.VAF) AS avg_vaf
4 FROM Sample_Mutation
5 GROUP BY Sample_Mutation.Sample_ID
6 HAVING AVG(Sample_Mutation.VAF) > 0.2
7 ORDER BY avg_vaf DESC;

```

## Rationale:

This query calculates the average VAF for each sample to capture the overall dominance of mutations within that sample. It uses GROUP BY to produce one value per sample, then applies HAVING to keep only samples with an average VAF above 0.2. The ORDER BY clause ranks these samples from highest to lowest average VAF, making it easy to spot the most dominant mutation profiles.

This approach summarizes mutation behavior at the sample level instead of evaluating individual mutations, providing a clearer and more comparable view across samples.

## Output:

| Sample_ID       | avg_vaf ▾ 1 |
|-----------------|-------------|
| TCGA-KT-A7W1-01 | 0.483091    |
| TCGA-E1-A7YY-01 | 0.468750    |
| TCGA-QH-A6XA-01 | 0.444615    |
| TCGA-DU-7299-01 | 0.429143    |
| TCGA-RY-A843-01 | 0.426087    |
| TCGA-S9-A6TW-01 | 0.415000    |
| TCGA-P5-A735-01 | 0.411667    |
| TCGA-TM-A7CF-01 | 0.402727    |
| TCGA-DU-7015-01 | 0.401000    |
| TCGA-HT-7470-01 | 0.393111    |

## 13. SQL:

```

1 SELECT
2     Copy_Number_Alteration.Sample_ID,
3     COUNT(*) AS amplification_count
4 FROM Copy_Number_Alteration
5 WHERE Copy_Number_Alteration.CNA_Value = 2
6 GROUP BY Copy_Number_Alteration.Sample_ID
7 ORDER BY amplification_count DESC;

```

## Rationale:

This query filters the data to keep only gene amplifications, then counts how many amplified genes exist in each sample. It uses GROUP BY to produce one count per sample and ORDER BY to rank samples from highest to lowest amplification burden.

This keeps the logic simple and avoids unnecessary joins since all required data is already in the CNA table. It also scales well and gives a clear way to compare samples and quickly identify those with the highest level of genomic amplification.

## Output:

| Sample_ID       | amplification_count ▾ 1 |
|-----------------|-------------------------|
| TCGA-TQ-A7RR-01 | 939                     |
| TCGA-HW-A5KM-01 | 759                     |
| TCGA-DH-5142-01 | 637                     |
| TCGA-IK-7675-01 | 592                     |
| TCGA-TM-A84F-01 | 541                     |

## 14. SQL:

```

1 SELECT Patient.Patient_ID, Survival.Overall_Survival_Months
2 FROM Patient
3 JOIN Diagnosis ON Patient.Patient_ID = Diagnosis.Patient_ID
4 JOIN Survival ON Diagnosis.Diagnosis_ID = Survival.Diagnosis_ID
5 ORDER BY Survival.Overall_Survival_Months DESC
6 LIMIT 5;

```

## Rationale:

This query connects patients to their survival data through the Diagnosis table to ensure each survival record is correctly assigned. It then orders patients by survival time from highest to lowest and selects the top 5 using LIMIT.

This approach is simple and efficient, providing a direct ranking of patients based on survival while preserving correct relationships between tables. It produces reliable results that support comparison across patients and help highlight cases with extended survival.

## Output:

| Patient_ID   | Overall_Survival_Months |
|--------------|-------------------------|
| TCGA-DU-6392 | 211.16                  |
| TCGA-DU-5870 | 182.33                  |
| TCGA-HT-7483 | 172.77                  |
| TCGA-S9-A7R1 | 169.84                  |
| TCGA-DU-7008 | 156.23                  |

## 9. Graph-Based Mutation Analysis Using Neo4j

This section presents the graph-based analysis of mutation data using Neo4j, following an initial SQL-based preprocessing pipeline. The objective is to identify biologically relevant genes associated with low-grade glioma (LGG) by integrating mutation frequency, functional impact, and subtype-specific patterns.

The analysis begins with characterizing the dataset in terms of IDH mutation status, followed by identifying recurrent genes across samples. These filtered genes are then used to construct a graph representation, allowing for the exploration and visualization of relationships between samples and genes.

This approach identifies a more intuitive understanding of mutation patterns and highlights key genes that distinguish between IDH-mutant and IDH-wildtype subtypes.

## Step 1: Distribution of samples by IDH status

```

1 SELECT
2     IDH_Status,
3     COUNT(*) AS sample_count
4 FROM Sample
5 GROUP BY IDH_Status;

```

| IDH_Status | sample_count |
|------------|--------------|
| Mutant     | 411          |
| Wildtype   | 88           |

To understand the composition of the dataset, a query was performed to count the number of samples in each IDH subtype. This step provides a baseline view of the cohort before conducting mutation-based analyses.

The results show that the dataset is dominated by IDH-mutant samples (411 samples), compared to a smaller number of IDH-wildtype samples (88 samples). This imbalance is important to consider in downstream analyses, particularly when comparing mutation frequencies between subtypes, as normalization or frequency-based comparisons are required to avoid bias.



## Step 2: Identification of recurrently mutated genes

```

1 SELECT Hugo_Symbol, COUNT(*) AS sample_count
2 FROM Sample_Mutation
3 JOIN Mutation ON Sample_Mutation.Mutation_ID =
  Mutation.Mutation_ID
4 JOIN Gene ON Mutation.Gene_ID = Gene.Gene_ID
5 GROUP BY Hugo_Symbol
6 HAVING COUNT(*) >= 5
7 ORDER BY sample_count DESC;

```

| Hugo_Symbol | sample_count | ▼ 1 |
|-------------|--------------|-----|
| IDH1        | 390          |     |
| TP53        | 316          |     |
| ATRX        | 220          |     |
| TTN         | 162          |     |
| CIC         | 131          |     |
| MUC16       | 68           |     |
| NOTCH1      | 52           |     |
| FUBP1       | 52           |     |
| PIK3CA      | 47           |     |
| EGFR        | 45           |     |
| NF1         | 44           |     |
| OBSCN       | 39           |     |
| FLG         | 35           |     |
| HMCN1       | 35           |     |
| SYNE1       | 34           |     |
| SMARCA4     | 32           |     |
| MUC17       | 32           |     |
| FAT2        | 31           |     |
| LRP2        | 30           |     |
| RYR2        | 27           |     |
| PCLO        | 27           |     |
| PTEN        | 26           |     |
| DST         | 25           |     |
| PIK3R1      | 25           |     |
| LRP1B       | 25           |     |

To identify recurrently mutated genes across the cohort, a query was performed to count the number of samples in which each gene is mutated. Only genes observed in at least five distinct samples were retained, ensuring that rare or potentially non-relevant mutations were excluded from further consideration. The results highlighted several highly recurrent genes, most notably IDH1 (390 samples), TP53 (316 samples), ATRX (220 samples), and TTN (162 samples), indicating that these genes are commonly altered across the dataset and may play central roles in glioma biology. This filtering step focuses the analysis on consistently mutated genes with higher biological relevance, providing a robust foundation for downstream investigations such as mutation impact assessment and subtype-specific characterization.

## Step 3: Filtering genes based on mutation impact

```

1 SELECT
2     Hugo_Symbol,
3     COUNT(DISTINCT Sample_ID) AS sample_count,
4     SUM(Primary_Consequence IN ('stop_gained','frameshift_variant')) AS truncating_events,
5     SUM(Primary_Consequence LIKE 'splice%') AS splice_events,
6     SUM(Primary_Consequence = 'missense_variant') AS missense_events
7 FROM Sample_Mutation
8 JOIN Mutation USING (Mutation_ID)
9 JOIN Gene USING (Gene_ID)
10 GROUP BY Hugo_Symbol
11 HAVING sample_count >= 5
12 ORDER BY truncating_events DESC;

```

| Hugo_Symbol | sample_count | truncating_events | splice_events | missense_events |
|-------------|--------------|-------------------|---------------|-----------------|
| ATRX        | 144          | 156               | 14            | 37              |
| CIC         | 93           | 58                | 4             | 59              |
| TP53        | 189          | 43                | 14            | 222             |
| FUBP1       | 44           | 35                | 9             | 2               |
| NF1         | 27           | 29                | 7             | 8               |
| ARID1A      | 19           | 18                | 0             | 4               |
| TTN         | 79           | 13                | 1             | 98              |
| NIPBL       | 15           | 12                | 2             | 8               |
| TCF12       | 14           | 12                | 2             | 0               |
| ZNF292      | 12           | 11                | 0             | 6               |
| BCOR        | 19           | 9                 | 0             | 6               |
| NOTCH1      | 37           | 8                 | 3             | 21              |
| ARID2       | 9            | 7                 | 0             | 13              |
| PTEN        | 21           | 7                 | 1             | 13              |
| SETD2       | 10           | 6                 | 0             | 6               |
| ARID1B      | 7            | 5                 | 0             | 8               |
| ASPM        | 8            | 5                 | 0             | 7               |
| HEATR5B     | 7            | 5                 | 0             | 4               |
| MYT1        | 5            | 5                 | 0             | 0               |
| PBRM1       | 9            | 5                 | 0             | 4               |
| PIK3R1      | 23           | 5                 | 2             | 7               |
| RBBP6       | 6            | 5                 | 0             | 3               |
| OBSCN       | 22           | 4                 | 0             | 20              |
| PTPRZ1      | 8            | 4                 | 0             | 6               |
| ANKRD17     | 9            | 3                 | 3             | 0               |

To further refine the set of recurrently mutated genes, an additional filtering step was applied based on the predicted functional impact of mutations. Synonymous and 3' UTR variants were excluded, as they are less likely to affect protein function directly and are often considered biologically neutral.

For each gene, the number of truncating events (stop-gained and frameshift variants), splice events, and missense mutations was quantified. A threshold of at least five truncating events was then applied to prioritize genes with a higher likelihood of functional disruption.

This analysis identified several genes with strong damaging mutation profiles, including ATRX (156 truncating events), CIC (58), TP53 (43), FUBP1 (35), and NF1 (29). These genes are not only recurrent across multiple samples but also exhibit mutation patterns consistent with significant loss of function or structural alteration.

By focusing on genes enriched in high-impact mutations, this step reduces the contribution of passenger mutations. It highlights candidates that are more likely to play a central role in glioma pathogenesis and disease progression.

#### Step 4: Comparison of selected genes across IDH subtypes

```

1 SELECT
2     IDH_Status,
3     Hugo_Symbol,
4     COUNT(*) AS sample_count
5 FROM Sample
6 JOIN Sample_Mutation USING (Sample_ID)
7 JOIN Mutation USING (Mutation_ID)
8 JOIN Gene USING (Gene_ID)
9 WHERE Hugo_Symbol IN ('TP53', 'ATRX', 'CIC', 'FUBP1')
10 GROUP BY IDH_Status, Hugo_Symbol;
```

| IDH_Status | Hugo_Symbol | sample_count |
|------------|-------------|--------------|
| Wildtype   | ATRX        | 70           |
| Mutant     | ATRX        | 144          |
| Mutant     | CIC         | 99           |
| Wildtype   | CIC         | 27           |
| Mutant     | TP53        | 216          |
| Wildtype   | TP53        | 75           |
| Mutant     | FUBP1       | 35           |
| Wildtype   | FUBP1       | 16           |

To evaluate whether the selected genes are associated with specific glioma subtypes, the distribution of mutations was compared between IDH-mutant and IDH-wildtype samples. For each gene, the number of distinct samples harboring mutations was calculated within each subtype.

The results reveal a strong subtype-specific pattern. ATRX is enriched in IDH-mutant samples (144 vs. 70), indicating a clear association with this subtype. A similar trend is observed for CIC (99 vs. 27) and FUBP1 (35 vs. 16), both of which are more frequently mutated in IDH-mutant tumors. TP53, while also

more common in IDH-mutant samples (216 vs. 75), shows a broader distribution and appears across both subtypes, reflecting its role as a more general tumor suppressor.

Overall, these findings demonstrate that several recurrently mutated genes, particularly ATRX, CIC, and FUBP1, are enriched in IDH-mutant gliomas, highlighting their potential as subtype-specific markers. This reinforces the biological distinction between IDH-mutant and IDH-wildtype tumors and supports the use of these genes in molecular classification and downstream analyses.

### Step 5: Frequency-based comparison across IDH subtypes

```

1 SELECT
2     Hugo_Symbol,
3     IDH_Status,
4     COUNT(*) AS mutated_samples,
5     COUNT(*) * 1.0 / total_samples AS frequency
6 FROM Sample
7 JOIN Sample_Mutation USING (Sample_ID)
8 JOIN Mutation USING (Mutation_ID)
9 JOIN Gene USING (Gene_ID)
10
11 JOIN (
12     SELECT IDH_Status, COUNT(*) AS total_samples
13     FROM Sample
14     GROUP BY IDH_Status
15 ) t USING (IDH_Status)
16
17 WHERE Hugo_Symbol IN ('ATRX', 'CIC', 'TP53', 'FUBP1')
18 GROUP BY Hugo_Symbol, IDH_Status;

```

| Hugo_Symbol | IDH_Status | mutated_samples | frequency |
|-------------|------------|-----------------|-----------|
| ATRX        | Wildtype   | 70              | 0.79545   |
| ATRX        | Mutant     | 144             | 0.35036   |
| CIC         | Mutant     | 99              | 0.24088   |
| CIC         | Wildtype   | 27              | 0.30682   |
| TP53        | Mutant     | 216             | 0.52555   |
| TP53        | Wildtype   | 75              | 0.85227   |
| FUBP1       | Mutant     | 35              | 0.08516   |
| FUBP1       | Wildtype   | 16              | 0.18182   |

To account for differences in sample size between IDH-mutant and IDH-wildtype groups, mutation frequencies were calculated for each gene within each subtype by normalizing the number of mutated samples to the total number of samples in that subtype.

The results show clear subtype-specific patterns. ATRX exhibits a higher frequency in IDH-wildtype samples compared to IDH-mutant samples (0.79545 vs. 0.35036), indicating a stronger relative enrichment in the wildtype group. A similar trend is observed for CIC (0.30682 vs. 0.24088) and FUBP1 (0.18182 vs. 0.08516), both of which show higher frequencies in IDH-wildtype tumors despite their higher absolute counts in the mutant group. TP53 also displays a higher frequency in IDH-wildtype samples (0.85227 vs. 0.52555), while remaining prevalent in both subtypes, confirming its broader role across glioma types.

These findings show that normalization by sample size changes the interpretation of enrichment patterns. The higher frequencies in IDH-wildtype samples indicate that mutations in ATRX, CIC, FUBP1, and TP53 are proportionally more common in this subgroup, even if absolute mutation counts are higher in IDH-mutant samples. This highlights the importance of frequency-based analysis when comparing groups of unequal size and provides a more accurate view of subtype-specific mutation burden.

## Step 6: Quantifying subtype-specific enrichment

```

1 SELECT
2     Hugo_Symbol,
3     SUM(IDH_Status = 'Mutant') / COUNT(*) AS mutant_freq,
4     SUM(IDH_Status = 'Wildtype') / COUNT(*) AS
5     wildtype_freq,
6     (SUM(IDH_Status = 'Mutant') / COUNT(*) /
7     NULLIF(SUM(IDH_Status = 'Wildtype') / COUNT(*), 0) AS
8     enrichment_ratio
9 FROM Sample
10 JOIN Sample_Mutation USING (Sample_ID)
11 JOIN Mutation USING (Mutation_ID)
12 JOIN Gene USING (Gene_ID)
13 WHERE Hugo_Symbol IN ('ATRX', 'CIC', 'TP53', 'FUBP1')
14 GROUP BY Hugo_Symbol;

```

| Hugo_Symbol | mutant_freq | wildtype_freq | enrichment_ratio |
|-------------|-------------|---------------|------------------|
| ATRX        | 0.6729      | 0.3271        | 2.05716906       |
| CIC         | 0.7857      | 0.2143        | 3.66635558       |
| TP53        | 0.7423      | 0.2577        | 2.88048118       |
| FUBP1       | 0.6863      | 0.3137        | 2.18775901       |

To further quantify the relationship between gene mutations and IDH subtypes, an enrichment ratio was calculated for each gene. This ratio represents the mutation frequency in IDH-mutant samples relative to that in IDH-wildtype samples, providing a direct measure of subtype-specific enrichment. The results reveal strong and consistent enrichment in IDH-mutant tumors. CIC shows the highest enrichment ratio (~3.67), followed by TP53 (~2.88), indicating that these genes are substantially more frequent in the IDH-mutant group. FUBP1 (~2.19) and ATRX (~2.06) also display clear enrichment, although they remain present at lower frequencies in IDH-wildtype samples.

Notably, all selected genes exhibit enrichment ratios greater than 1, confirming a clear bias toward IDH-mutant samples. This indicates that these mutations are not randomly distributed but are strongly associated with a specific molecular subtype.

These findings provide robust quantitative evidence of subtype specificity, highlighting ATRX, CIC, FUBP1, and TP53 as key markers of IDH-mutant gliomas and supporting their relevance for downstream analysis and biological interpretation.

## Step 7: Preparing data for graph-based analysis

```

1 SELECT DISTINCT
2     Sample_ID,
3     IDH_Status,
4     Codeletion_Status,
5     Hugo_Symbol
6 FROM Sample
7 JOIN Sample_Mutation USING (Sample_ID)
8 JOIN Mutation USING (Mutation_ID)
9 JOIN Gene USING (Gene_ID)
10 WHERE Hugo_Symbol IN ('ATRX', 'CIC', 'TP53', 'FUBP1')
11 ORDER BY Sample_ID;

```

| Sample_ID       | IDH_Status | Codeletion_Status | Hugo_Symbol |
|-----------------|------------|-------------------|-------------|
| TCGA-CS-4938-01 | Mutant     | Non-Codeleted     | TP53        |
| TCGA-CS-4941-01 | Wildtype   | NULL              | ATRX        |
| TCGA-CS-4942-01 | Mutant     | Non-Codeleted     | TP53        |
| TCGA-CS-4943-01 | Mutant     | Non-Codeleted     | ATRX        |
| TCGA-CS-4943-01 | Mutant     | Non-Codeleted     | TP53        |
| TCGA-CS-4944-01 | Mutant     | Non-Codeleted     | ATRX        |
| TCGA-CS-5390-01 | Mutant     | Codeleted         | FUBP1       |

To prepare the data for graph construction in Neo4j, a final integrated dataset was generated linking each sample to its mutated genes while retaining key clinical attributes, including IDH status and codeletion status.

The resulting table represents a clear sample–gene relationship, where each row corresponds to a mutation event connecting a specific sample to a gene. This structure directly supports graph modeling, with samples and genes represented as nodes and their relationships captured as edges.

By including IDH status and codeletion status, each sample node carries clinically relevant annotations, enabling the visualization of subtype-specific patterns and facilitating stratified analysis within the network.

This dataset forms the final step in data preparation, providing a clean and structured input for Neo4j to construct and explore the tumor mutation network.

## Step 8: Graph Construction and Data Transformation in Neo4j

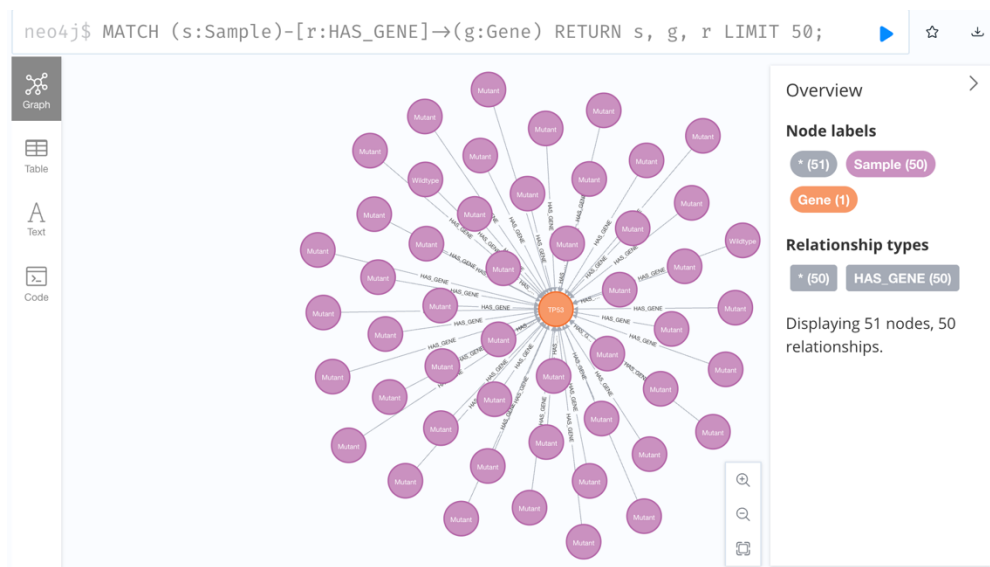
Following the preparation of the final integrated dataset, the results were exported into a CSV file named “*For Neo4j.csv*”. This file contains distinct sample–gene relationships along with key clinical attributes, including IDH status and codeletion status.

To construct the graph database, a custom Python script was developed to transform the tabular data into executable Cypher queries. The script reads the CSV file, performs essential preprocessing steps such as trimming whitespace, standardizing data types, and removing incomplete records, and then converts each row into structured Cypher statements.

Within this process, each sample is represented as a node labeled *Sample*, uniquely identified by *Sample\_ID*, with associated properties for *idh\_status* and *codeletion\_status*. Each gene is represented as a node labeled *Gene*, identified by *Hugo\_Symbol*. The relationship between samples and genes is explicitly modeled using directed edges (*HAS\_GENE*), where each edge represents a mutation event linking a sample to a gene.

This approach avoids direct CSV ingestion into Neo4j, providing greater control over data cleaning and formatting while preventing common parsing errors related to improperly formatted fields. The generated Cypher script was then executed to populate the Neo4j database, resulting in a graph structure that accurately captures the relationships between samples and their mutated genes, enabling efficient network-based analysis and visualization.

## Step 9: Visualization



```
neo4j$ MATCH (s:Sample) WHERE s.idh_status = "Mutant" SET s:Mutant;
```

```
neo4j$ MATCH (s:Sample) WHERE s.idh_status = "Wildtype" SET s:Wildtype;
```

```
neo4j$ MATCH (s:Sample) WHERE s.codeletion_status = "Codeleted" SET s:Codelet...
```

```
neo4j$ MATCH (s:Sample) WHERE s.codeletion_status = "Non-Codeleted" SET s:Non...
```

The initial graph visualization displayed the connections between samples and the selected genes. However, all sample nodes appeared visually similar, making it difficult to distinguish biological subtypes or extract meaningful patterns directly from the graph.

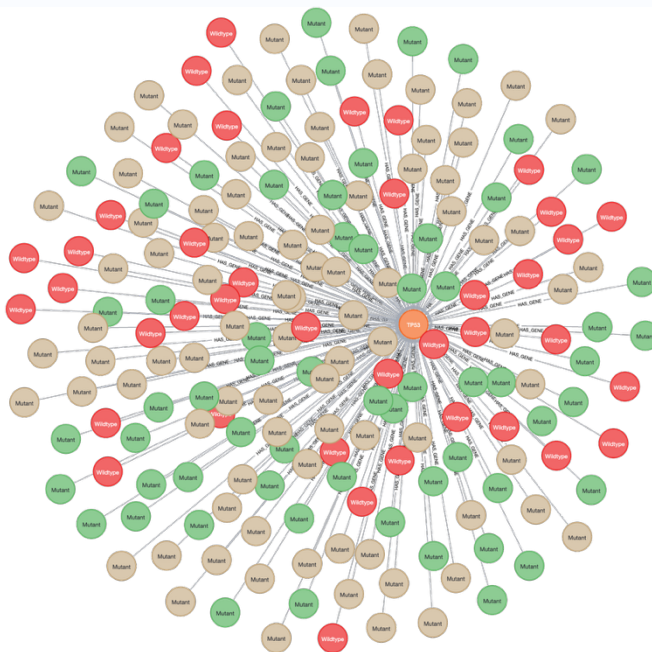
To improve interpretability, node labeling was introduced based on IDH status. Samples were categorized into two groups: IDH-mutant and IDH-wildtype. This modification allowed visual separation of samples and enabled clearer identification of subtype-specific clustering patterns.

Following this, an additional refinement was applied by incorporating codeletion status. Samples were further classified into codeleted and non-codeleted groups. This added a second layer of biological information, allowing simultaneous visualization of both IDH status and co-deletion alteration patterns.

These incremental modifications significantly improved the clarity of the graph. Instead of a uniform structure, the network began to reveal meaningful groupings, where clusters of samples sharing similar biological characteristics became visually apparent.

### TP53 Sample Interaction Network

```
neo4j$ MATCH (s:Sample)-[:HAS_GENE]-(g:Gene {name: "TP53"}) RETURN s, g;
```



#### Overview

#### Node labels

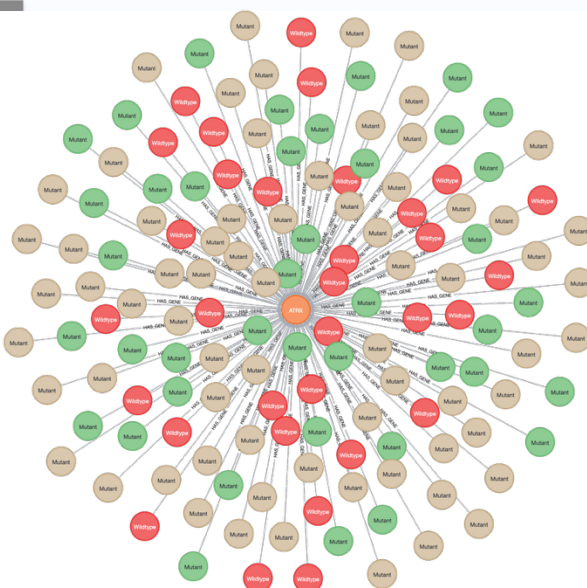
- \* (190) Codeleted (61)
- Mutant (149)
- Sample (189) Gene (1)
- NonCodeleted (88)
- Wildtype (40)

Graph showing 189 samples connected to TP53, with nodes colored according to IDH mutation status and 1p/19q codeletion status. The majority of samples are IDH-mutant, while a smaller fraction represents IDH-wildtype cases. Within the IDH-mutant group, a considerable proportion is non-codeleted, with a smaller subset exhibiting 1p/19q codeletion.

This distribution indicates that TP53 alterations are predominantly associated with IDH-mutant tumors, particularly within the non-codeleted subgroup. The visualization clearly reflects the known biological pattern in which TP53 mutations are enriched in IDH-mutant, non-codeleted gliomas, while being less frequent in codeleted or wild-type cases.

### ATRX Sample Interaction Network

```
neo4j$ MATCH (s:Sample)-[:HAS_GENE]-(g:Gene {name: "ATRX"}) RETURN s, g;
```



#### Overview

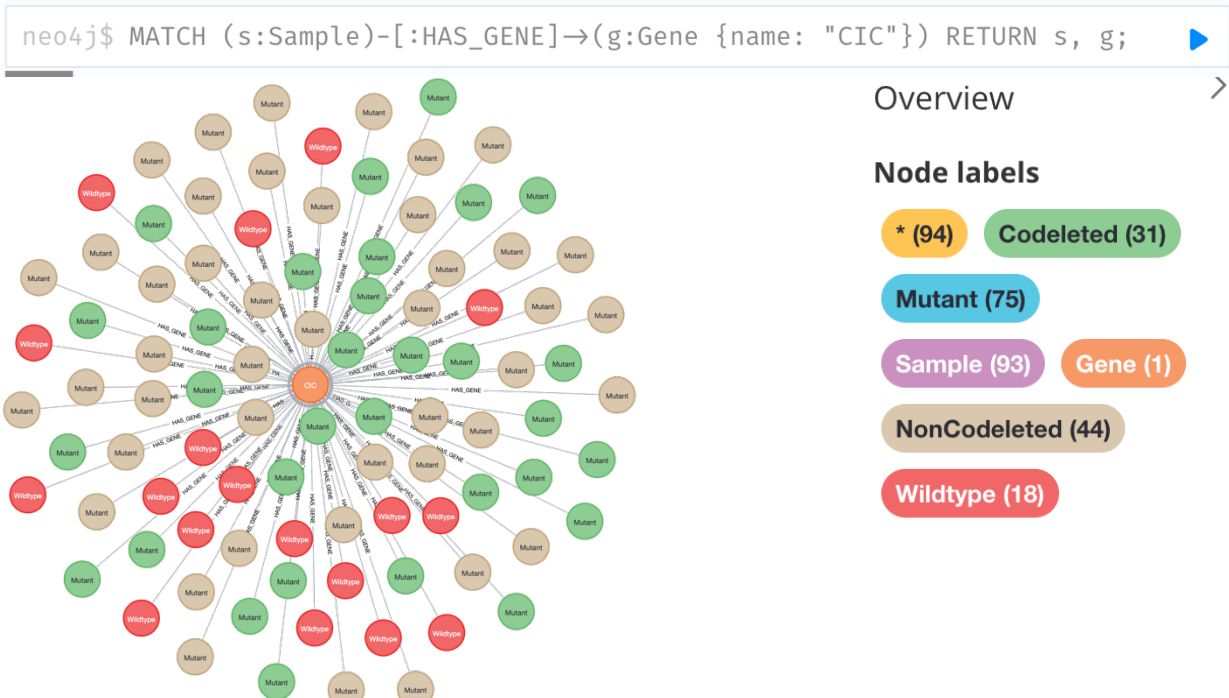
#### Node labels

- \* (144) Sample (143)
- Wildtype (32) Gene (1)
- Mutant (111)
- NonCodeleted (70)
- Codeleted (41)



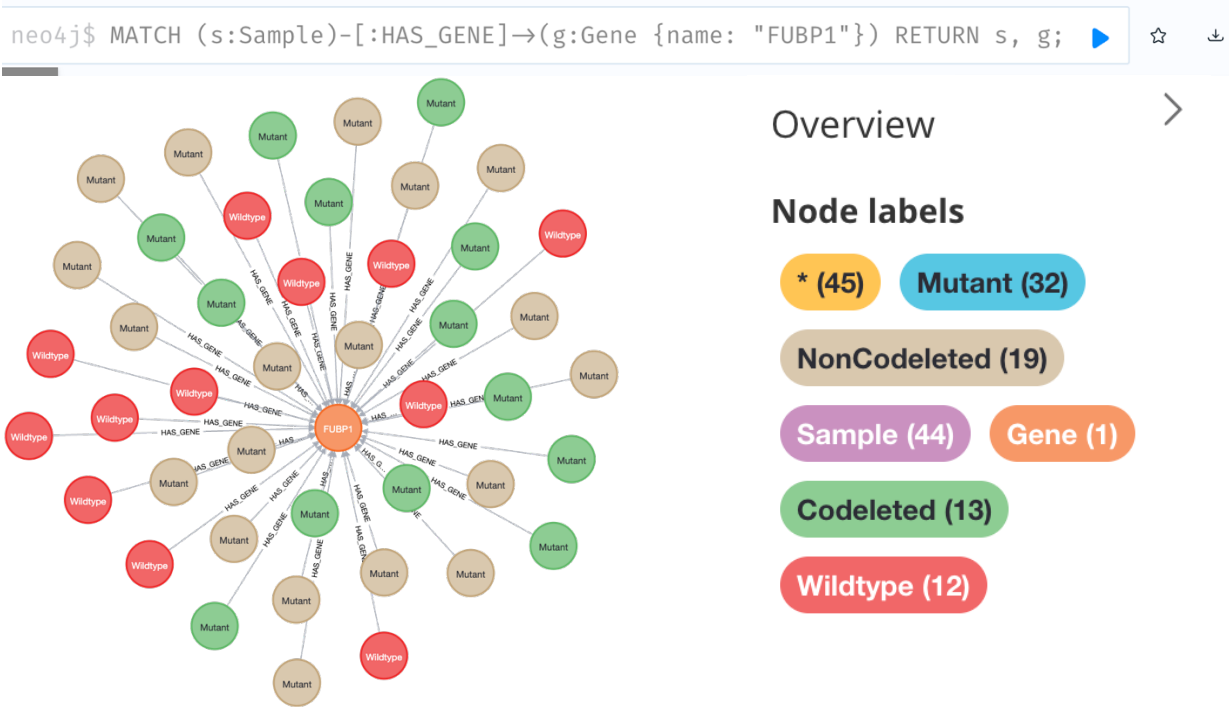
Graph showing 143 samples connected to ATRX, with nodes colored according to IDH mutation status and 1p/19q codeletion status. The majority of samples are IDH-mutant, while a smaller subset represents IDH-wildtype cases. Within the IDH-mutant group, most samples are non-codeleted, with only a limited number of codeleted cases. This distribution is consistent with the established biological association of ATRX alterations with IDH-mutant, non-codeleted gliomas. The visualization highlights the strong enrichment of ATRX mutations in this subgroup, supporting its characteristic role in astrocytoma rather than oligodendroglioma.

CIC Sample Interaction Network



Graph showing 93 samples connected to CIC, with nodes colored according to IDH mutation status and 1p/19q codeletion status. The majority of samples are IDH-mutant, with a substantial proportion belonging to the codeleted group, while a smaller fraction represents wildtype and non-codeleted cases. This distribution reflects the well-established association of CIC alterations with IDH-mutant, codeleted gliomas. The visualization highlights the enrichment of CIC mutations within the codeleted subgroup, supporting its characteristic role in oligodendroglioma and distinguishing it from non-codeleted or wild-type tumor profiles.

FUBP1 Sample Interaction Network





Graph showing 44 samples connected to FUBP1, with nodes colored according to IDH mutation status and 1p/19q codeletion status. The majority of samples are IDH-mutant, with a notable proportion belonging to the codeleted group, while a smaller subset represents wildtype and non-codeleted cases.

This distribution aligns with the established association of FUBP1 alterations with IDH-mutant, codeleted gliomas. The visualization highlights the enrichment of FUBP1 mutations within the codeleted subgroup, supporting its characteristic role in oligodendroglioma and its relative scarcity in non-codeleted or IDH-wildtype tumors.

### 10. Integrated table of key mutated genes with functional, clinical, and frequency annotations from multiple databases

Table 3. Integrated Functional and Clinical Annotation of Key Mutated Genes in LGG (Reference 1-13)

| Gene  | Function (GeneCards)                                              | Variants (COSMIC / cBioPortal)                                             | Clinical Significance (OncoKB / ClinVar)                                                         | Population Frequency (LGG – cBioPortal) | Molecular Subtype Association                    | Notes (TCGA LGG / cBioPortal)                                                                      |
|-------|-------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------|
| TP53  | Tumor suppressor regulating cell cycle, DNA repair, and apoptosis | Predominantly missense mutations; common hotspots include R175, R248, R273 | Well-established tumor suppressor driver; associated with genomic instability and poor prognosis | ~ 45–50%                                | Astrocytoma (IDH-mutant)                         | Strong co-occurrence with ATRX; largely mutually exclusive with CIC/FUBP1 and 1p/19q codeletion    |
| ATRX  | Chromatin remodeling; telomere maintenance via ALT pathway        | Truncating, frameshift, loss-of-function mutations                         | Tumor suppressor; defines astrocytoma lineage; associated with ALT activation                    | ~35–45%                                 | Astrocytoma (IDH-mutant)                         | Strong co-occurrence with TP53; rare in oligodendroglioma                                          |
| CIC   | Transcriptional repressor regulating MAPK signaling               | Missense and truncating mutations, often affecting HMG-box domain          | Tumor suppressor; associated with oligodendroglioma; favorable prognosis                         | ~15–25%                                 | Oligodendroglioma (IDH-mutant, 1p/19q codeleted) | Strong association with 1p/19q codeletion; co-mutated with FUBP1; largely exclusive with TP53/ATRX |
| FUBP1 | Transcriptional regulator including MYC modulation                | Frameshift, truncating mutations                                           | Tumor suppressor; contributes to oligodendroglioma development                                   | ~8–12%                                  | Oligodendroglioma (IDH-mutant, 1p/19q codeleted) | Frequently co-mutated with CIC; enriched in codeleted tumors                                       |

### 11. Reproduction Instructions

The database can be fully rebuilt from raw data using a two-stage workflow. Data cleaning and harmonization were performed locally on a Mac environment, while database construction, population, and querying were executed inside a virtual machine to ensure consistency and reproducibility.

#### Local Environment (Mac laptop)

- Python 3.13
- Python packages: pandas, numpy, pathlib

#### Used for:

- Data cleaning
- Data validation
- Sample-level harmonization
- Gene-level harmonization
- Final dataset generation
- SQL generation for database population

#### Virtual Machine Environment

- VirtualBox 7.2.0
- Pre-configured Linux virtual machine
- MySQL
- Neo4j

#### Used for:

- Database creation
- Database population
- SQL querying
- Graph-based analysis

#### Setup

1. Install VirtualBox and import the provided virtual machine image
2. Start the virtual machine and log in using:

- Username: xxxxx (encrypted for confidentiality)
- Password: xxxxx (encrypted for confidentiality)
- 3. Create a shared folder named: DB-Final
- 4. Access the shared folder inside the virtual machine at: /media/sf\_DB-Final/
- 5. Confirm MySQL access inside the virtual machine (download if needed using % sudo dnf install mysql)

**Workflow**

1. Run all data cleaning and harmonization scripts locally on the Mac  
Execution sequence:  
From steps 1-7 can be run all together using run\_pipeline.py
  1. Run data cleaning scripts:
    - o Clinical and sample cleaning (i.e, 01\_clean\_clinical\_sample.py)
    - o Mutation cleaning (i.e, 02\_clean\_cna.py)
    - o CNA transformation and cleaning (i.e, 03\_clean\_expression.py)
    - o Gene expression transformation (i.e, 04\_clean\_mutations.py)
  2. Run validation (i.e, 05\_validator.py):
    - o Check required fields
    - o Remove invalid or inconsistent records
  3. Run sample harmonization (i.e, 06\_sample\_harmonization.py):
    - o Retain only samples present across all datasets
  4. Run gene harmonization (i.e, 07\_gene\_harmonization.py):
    - o Retain only shared genes across mutation, CNA, and expression
  5. Run gene identifier mapping (i.e, 08\_hgnc\_mapping.py):
    - o Standardize gene identifiers using the HGNC reference
  6. Run final harmonization (i.e, 09\_finalize.py):
    - o Generate fully matched datasets across all omics layers
  7. Run validation report (i.e, 10\_report.py):
    - o Confirm counts of patients, samples, genes, and mutations
    - o Will help confirm whether the population step is complete or not.
  8. Generate SQL population script (i.e, SQL\_generation\_for\_populating\_the\_data.py):
    - o Convert cleaned datasets into structured INSERT statements
    - o Step 8 is run separately after run\_pipeline.py.
2. Transfer SQL scripts (i.e, 01\_create\_database\_schema.sql, 02\_load\_cleaned\_data.sql) to the virtual machine using the shared folder
  - o 1st file is to make the database
  - o 2nd file is to populate the database
3. Build and populate the database inside the virtual machine
  - o To build the database schema, run: mysql -u root -p < 01\_create\_database\_schema.sql
  - o To populate the database with the cleaned and harmonized data, run: mysql -u root -p Database < 02\_load\_cleaned\_data.sql
  - o To export the populated database as a MySQL dump file to the shared document, run: mysqldump --single-transaction -h 127.0.0.1 -P 3306 -u root -p Database > 03\_mydump.sql. Then copy the file and paste it in your shared file.
4. Execute the demonstrated SQL queries on the database
  - o Using the output of the last few SQLs (pages 20-25), export the Neo4j.csv file.
  - o Use For Neo4j.py to turn the CSV into a Neo4j Cypher import script
5. Perform Neo4j analysis (Pages 25-28)
6. Look up the genes of interest, including TP53, ATRX, CIC, and FUBP1, in major bioinformatics and cancer genomics databases such as GeneCards, COSMIC, cBioPortal, OncoKB, and ClinVar to obtain information on gene function, mutation patterns, clinical relevance, population frequency, and molecular subtype associations.

**12. Limitations**

The current database design focuses on gene-level integration of mutation, copy number alteration, gene expression, and selected clinical variables. This creates a clean, consistent, and highly interpretable structure, but it also introduces several limitations. The dataset is restricted to a single cancer type, Low-Grade Glioma. While the schema is designed to be scalable, extending it to multiple cancer types will require additional validation to handle increased heterogeneity in both clinical and molecular data. Several omics layers were intentionally excluded, including methylation, proteomics, and structural variants. Each of these data types requires distinct preprocessing pipelines and more complex data models. Their exclusion limits the ability to perform fully integrated, multi-layer biological analysis.

The database relies on preprocessed datasets rather than raw sequencing data. This improves consistency and usability, but introduces dependency on upstream processing methods and reduces flexibility in redefining analytical workflows.

Some derived attributes, such as Variant Allele Frequency (VAF), are stored directly rather than computed dynamically. This improves query performance and simplifies downstream analysis, but reduces flexibility if alternative calculation strategies are required.

Clinical variables related to treatment, longitudinal timelines, and staging were excluded to maintain a focused gene-centered structure. As a result, the database does not currently support treatment-response analysis or time-dependent clinical modeling.

### 13. *Future Work*

Several extensions can improve the scope and analytical power of the database:

- Integrate additional omics layers, such as methylation and proteomics
- Extend the schema to support multiple cancer types
- Incorporate treatment data and longitudinal clinical variables
- Enable dynamic computation of derived attributes such as VAF
- Integrate advanced analytical frameworks, including machine learning models

These improvements would transform the database from a structured integration platform into a more comprehensive system for biological discovery, predictive modeling, and clinical insight.

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